

Earnings targets, strategic patent sales, and patent trolls[☆]Jinhwan Kim^{a,*} , Kristen Valentine^b ^a Stanford Graduate School of Business, Stanford University, USA^b Terry College of Business, The University of Georgia, USA

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ABSTRACT

Innovative public firms sell significantly more patents in the last month of their fiscal year compared to earlier in the year. Consistent with reporting incentives driving these sales, final month patent sales are disproportionately internally generated, non-core patents with high potential accounting gains. Strategic patent sales are more pronounced among firms with incentives to meet earnings expectations – especially in periods when firms narrowly beat expectations and when executive incentives predominate. Consistent with these incentives, managers engage in high insider equity sales following strategic patent sales. In contrast, final month patent sales are less likely for private firms. Patents sold in the last month are litigated more frequently because they are disproportionately sold to “patent trolls”, who opportunistically acquire patents to engage in litigation. We highlight a novel consequence and externality of corporate reporting incentives: its contribution to strategic patent sales, which in turn impacts the market for innovation and litigation.

1. Introduction

Patent protection is a societal grand bargain, where an inventor receives temporary exclusivity rights in exchange for disclosure. An important, but often overlooked, fact about patents is that they are frequently bought and sold, and the market for patents is economically important. Industry reports suggest that the total asking price for patents on the market was about \$40 billion in 2022 and prior research finds that 29 % of granted patents are ultimately sold at least once (Richardson et al. (2023); Kim and Valentine (2023)). Given the economic significance of these transactions, understanding the motivations for when, and to what extent firms sell patents is central to academic research as well as in designing optimal policy. We posit that an important factor motivating firms to sell their patents is the incentive to boost reported profits to meet earnings expectations. Patent sales can increase short-term profits but

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could be suboptimal in terms of value creation. Our primary goal is to examine whether firms engage in strategic patent sales, and if so, the implications of these sales for the innovation environment more broadly.

To investigate whether firms use patent sales to manage earnings upward, we draw on two key observations. First, U.S. GAAP requirements can make selling patents an effective tool to manage reported earnings. Under GAAP, firms record a gain on the sale of patents when the proceeds from the sale exceed the patents' book value. However, GAAP does not allow firms to capitalize the significant costs of scientists' salaries and other R&D expenditures for internally developed patents, which translates into low book values. Thus, selling these patents likely results in greater gain recognition that boosts reported earnings. Second, because R&D investments are inherently uncertain, not all resulting discoveries will be equally valuable to the firm. Thus, firms will naturally consider selling the rights to patentable discoveries that are tangential to their core business. Over time, these non-core patents can accumulate into a reserve of patents the firm intends to sell eventually. We expect that firms can strategically time the sale of these patents when motivated by earnings management.¹

However, strategic patent sales could represent suboptimal use of the firm's innovative portfolio, whether via holding in reserve patents the firm intends to sell or selling patents that could be more profitably developed or licensed in-house. For this reason, we expect firms to exercise the "real option value" of selling patents to manage earnings such that firms wait until the end of a fiscal year to sell patents only if it becomes necessary (Oyer (1998); Kerstein and Rai (2007); Dechow et al. (2010); Das et al. (2011); Beardsley et al. (2021); Ertan (2022); Gong et al. (2023)). If the potential costs are sufficiently high, a manager may not be willing to strategically sell patents. If so, any elevated patent sales in the final month could be due to alternative explanations. For example, it is possible that firms sell patents at the end of the fiscal year as part of routine innovative portfolio rebalancing, for tax-motivated reasons, or due to financial constraints. Our empirical strategy is to document the timing of patent sales within a firm's fiscal year and then examine whether patent sales timing varies predictably with incentives to meet earnings expectations or varies for other reasons.

We create a novel database of public firms' patent sales provided by the U.S. Patent and Trademark Office ("USPTO") Patent Assignment data. The database allows us to observe the identity of the entities involved in a patent sale (e.g., public firms vs. "patent trolls") and when the patent transaction takes place.² Using this database, we construct a sample of firm-months from 1985 to 2021. Our primary research design regresses the number of patent sales on indicators for the months leading up to the firm's fiscal year end.³ After including firm-year and calendar month fixed effects, the indicator for the final month of a firm's fiscal year can be interpreted as the amount of "abnormal" patent sales occurring in the last month of the fiscal year ("month *M*") incremental to the firm's patents sold during the first six months of the year ("benchmark months").

Descriptively, we first examine the characteristics of patents sold in public firm's final fiscal month and compare those to the characteristics of patents sold at other times during the year or to patents retained in the firm. We find patents sold in the last fiscal month are more valuable, as proxied by forward citations, number of claims, and patent generality (Hall et al. (2001)). Intuitively, we also find the patents retained by the firm are more valuable than the patents sold in the final fiscal month. Despite patents sold in month *M* being more valuable, they are significantly *less likely* to have an associated 8-K disclosure (18.9 %) relative to patents sold in other months (32.9 %). The observed lower univariate rate of disclosure could be due to managers exercising their discretion to conceal their strategic intentions at the time of sale, or because firms are selling non-core patents they do not consider material.⁴

In our main empirical specification, we find that abnormal patent sales in month *M* are approximately 1.1 % greater on a per *month* basis, or 13.2 % annualized, which is economically material. This translates into roughly 2600 more aggregate patent sales over our sample period. This is consistent with firms strategically delaying the sale of patents, illustrating the real option value of patent sales. In contrast, the effects in the months leading up to month *M* are generally no different from the sales relative to the benchmark months.

Managers exercise joint control over both operating decisions and financial reporting choices (Roychowdhury et al. (2019); Huang

¹ Anecdotal evidence suggests patent sales can be used to achieve earnings targets. Unova, Inc., an industrial technology company, reported that it sold approximately 150 wireless local-area networking patents to Broadcom Corporation on December 24, 2002, for \$24 million, resulting in a gain which the firm said, "will positively impact UNOVA's 4th quarter and full-year 2002 earnings". Prior to year-end, analysts expected Unova to have a net loss per share of \$1.14. Unova's actual net loss was only \$0.82 per share. While the total reported gain recognized was likely somewhat less than the full \$24 million selling price, the selling price per share accounts for \$0.41, suggesting that this patent sale in the last month of their fiscal year likely moved the firm from missing to beating analyst expectations. While this saliently disclosed example demonstrates the plausibility of a patent sales strategy, in practice, relatively few patent transactions are transparently disclosed. We further develop this point in Section 4.1.2.

² "Patent trolls" are a pejorative term for entities termed non-practicing entities ("NPEs") or patent assertion entities. These entities do not conduct R&D or invent new products but rather purchase patents and initiate patent infringement lawsuits against operating companies. We use the phrase patent troll following the colloquial usage by politicians and firm managers or alternatively NPE interchangeably throughout the paper. See further discussion in Section 2.2.

³ While the U.S. Patent and Trademark Office (USPTO) assignment data tracks the number of patent sales, it does not include the transaction price. Accordingly, we are unable to use the dollar amount of patent sales to examine our research question.

⁴ As we discuss in Section 4.1.2, a firm's discretion in determining the materiality threshold could explain why we do not observe an 8-K filing for every patent sale. Our primary interest is in identifying the significantly lower rate of 8-K filings for month *M* patent sales relative to other months, *ceteris paribus*, which would be consistent with patent sales in month *M* being strategic. Also, investors are unlikely to observe patent sales via patent office disclosure in a timely manner. For a patent sale to be valid in court, the parties must record the patent assignment with the USPTO (Serrano (2010)). The USPTO maintains a patent assignment dataset that details the execution date for each patent sale as well as the date the transaction is reported to the patent office. Marco et al. (2015) report that on average, there is a lag of 182 days between the transaction's execution date and recording date. As researchers, we observe the execution date with hindsight, but investors at the time of patent sale may not be able to observe the sale in the USPTO data for months after the transaction.

et al. (2020)). In our setting, joint control means that if managers sell patents to achieve short-term gains motivated by earnings management, they also have incentives to use available reporting discretion to reduce the transparency of patent sales. Thus, to complement our main results, we further examine the disclosure of patent sales in an 8-K in a multivariate context and find that month *M* patent sales are significantly less likely to be disclosed through 8-K filings relative to patent sales occurring at other months during the year. This observation is consistent with the interpretation that managers use the available discretion in disclosure rules to obfuscate their strategic patent sales.

To examine whether patent sales at fiscal year-end are likely motivated by earnings management, we conduct three tests. First, we exploit variation in the types of patents sold, including whether the patent was internally developed versus acquired as well as whether the patent is core to the firm's operations versus non-core. With respect to internally developed versus acquired patents, we exploit their asymmetric accounting treatment. U.S. GAAP does not allow firms to capitalize the full fair value of internally developed patents but does allow firms to capitalize the fair value of purchased patents. Thus, the reported gain on sale is likely greater for internally developed patents and strategic motivations would suggest firms sell more internally developed than acquired patents at year end. Consistent with strategic motivations, we find public firms sell 13.9 % more internally developed patents than externally acquired patents in month *M*. Regarding core versus non-core patents, we find firms disproportionately sell non-core rather than core patents in the final month of the fiscal year, on average. This suggests that the average firm is likely strategically timing the sale of patents they intended to sell, rather than sacrificing rights to the firm's core technologies.⁵

Second, we exploit variation in a firm's incentive to meet or beat earnings expectations. Firms that *just* meet or beat earnings targets likely represent those with strong earnings management incentives (e.g., Bartov et al. (2002); Dhaliwal et al. (2004); Bhojraj et al. (2009); Dechow et al. (2010); Quinn (2018); Johnson et al. (2020)). We find that the abnormal patent sales effect in month *M* is 1.4 %–1.6 % higher per month in periods where a firm just meets or beats earnings expectations by 1 cent. Also, we plot histograms of patent sales across bins of earnings surprises, similar to Burgstahler and Dichev (1997), and find a stark discontinuous jump in the number of firms just above the threshold of meeting earnings targets versus the number of firms just below, consistent with strategic incentives motivating firms' patent sales.⁶ Moreover, the reduction in 8-K disclosure of patent sales is more pronounced for firms that just meet or beat earnings expectations, consistent with managers exercising joint control over both operating decisions and financial reporting choices (Roychowdhury et al. (2019); Huang et al. (2020)). This cross-sectional variation further supports the interpretation that managers deliberately reduce disclosure when patent sales are more likely to be driven by earnings management incentives.

Third, we exploit variation in the extent to which an executive's compensation incentivizes meeting earnings expectations. We document that the month *M* patent sales effect is 1.1 %–1.2 % higher per month in periods when the executives' compensation has greater sensitivity to earnings and stock prices (Leone et al. (2006)). Also, firms with high patent sales in the last month of the year have a 17.5 %–40.5 % increase in net insider equity sales the next quarter. Decomposing insider sales into routine versus abnormal trades following Cohen et al. (2012), we find that this increased trading is more pronounced in abnormal sales rather than pre-scheduled transactions. This pattern suggests that managers personally capitalize on the temporary earnings boost from strategic patent sales through informed trading, creating private benefits while potentially disadvantaging outside investors. The collective results from these three tests are consistent with managers strategically selling patents for earnings-motivated reasons.

We also examine the importance of alternative explanations, such as firms rebalancing their patent portfolio at year end, tax incentives, and financial constraints, in explaining the abnormal patent selling in month *M*. The results from these tests do not support these alternatives explaining our findings.⁷ Finally, our results are robust to excluding December year-end firms, various fixed effect structures, and using a Poisson model. Taken together, our findings are consistent with firms strategically selling patents in response to earnings pressure and are less consistent with alternative explanations.

We then explore the consequences of strategically motivated patent sales for the broader patent system. One important entity in the patent system is non-practicing entities (NPEs, also known as "patent trolls"). If patents sold to NPEs occur evenly throughout the year, approximately 8 % of sales (=1/12 months) would happen in the last month of the fiscal year. However, in our sample, 55 % of sales made to NPEs occur in the last fiscal month, suggesting patent trolls play an important role in facilitating month *M* patent transactions. We find that both the proportion of patents sold to NPEs relative to other buyers and the proportion of patent infringement cases asserted by NPEs relative to other plaintiffs increases for patents sold in month *M*. Importantly, this pattern of results is significantly more pronounced for firms that just meet or beat earnings expectations, providing compelling evidence that strategic earnings management motivations drive these transactions rather than other institutional factors. These findings are consistent with NPEs

⁵ Given the selective disclosure as noted in Section 4.1.2 of the paper and discussed above, we cannot obtain sales agreements for all patents transferred in our sample. Nonetheless, anecdotal evidence suggests firms can include in the sales agreement the provision that the seller retains the right to continue to use the patented technology royalty-free. In these instances, the seller is not giving up the ability to use their previously developed invention, but rather, is losing the right to license the underlying technology or sue for infringement directly.

⁶ To triangulate whether strategic patent sales have the potential to impact a firm's reported EPS, we hand collect the transaction amounts from eight identifiable examples where patent transaction values were disclosed in an 8-K. Across these examples, the median value per patent was \$876,190 and the median total transaction value was \$4,600,000, with approximately 5 patents trading in a given transaction. Using the average common shares of 103.95 million, we infer that patent sales can move EPS by four to five cents per share (\$4,600,000/103,950,000 shares), though we acknowledge that these identifiable examples may not be representative of the remaining population of patent sales.

⁷ Specifically, we perform two tests. First, we find that private firms are significantly less likely to sell patents in month *M* than public firms. This differential effect suggests our findings are incremental to institutional features that are common to both public and private firms (such as patent portfolio rebalancing and tax incentives). Second, financially constrained firms don't have more pronounced patent selling activity in month *M*, inconsistent with distress as an underlying mechanism.

strategically acquiring patents from firms that are likely selling their intellectual property under pressure to meet earnings targets and subsequently using these patents to initiate litigation against other companies for profit.⁸

We speak to the literature on the real effects of financial reporting (Leuz and Wysocki (2016)) by demonstrating that financial reporting incentives can distort the market for innovation and prompt patent sales to potentially unproductive users. Furthermore, the finding that firms sell more internally generated patents suggests that reporting rules that preclude firms from capitalizing the full cost of internally developed patents make selling patents an attractive earnings management tool and can have the unintended consequence of facilitating sales to patent trolls. We also contribute to the emerging literature at the intersection of accounting and innovation (e.g., Glaeser (2018); Kim and Valentine (2021); Kim and Valentine (2023); Glaeser and Lang (2024)) by expanding our understanding of the forces that shape the underexplored secondary patent market.

Our findings are also relevant to the literature on the market for patents, innovation, and NPEs. Serrano (2010) documents the determinants of patent transfers and finds that small innovators and private inventors are the parties most likely to sell their patents. We complement this evidence by providing a dataset that links patent sales to public firms and shows conditions under which public firms are more likely to sell patents (i.e., in the presence of financial reporting incentives). Hochberg et al. (2018) find that the patent market can facilitate lending to start-up firms given it provides lenders with a liquidation option for patents pledged as collateral. Importantly, we provide evidence that strategic patent selling, driven by financial reporting incentives, can facilitate NPE litigation, which can be detrimental to welfare (Cohen et al. (2019)).

Finally, we contribute to research on managers' use of real earnings management. Real earnings management studies most relevant to our setting examine the use of fixed asset sales to manage earnings (Bartov (1993)) and discretionary cuts to R&D expenditures (Bushee (1998); Kothari et al. (2016); Bereskin et al. (2018)). Because real actions must be taken within the fiscal period to effectively manage earnings and managers have better information about the degree of intervention necessary as the end of the period approaches, real earnings management via asset sales should occur closer to year end (Zang (2012)). While research on asset sales as an earnings management strategy has not been able to directly test the timing of sales (Bartov (1993)), we are able to demonstrate the timing of patent sales. We also contribute to the literature on discretionary cuts to R&D as an earnings management tool by highlighting the other side of the coin – managers may reduce R&D spending to manage earnings but can also sell R&D-related assets.

2. Background and hypothesis development

2.1. Patent market background

Since 2013, the cumulative patent market has an estimated total asking price of about \$40 billion, with the average asking price per patent of \$210,000 over the last five years (Richardson et al. (2023)). While the best use of the patent market should be to reallocate intellectual property to the most productive user (Galasso et al. (2013)), in practice, the patent market has been disaggregated and opaque. Historically, there has not been a central exchange with transparent pricing, and even with recent online marketplaces, almost 40 % of all patent assets listed for sale do not include even pricing information (Richardson et al. (2023)). The lack of a transparent, centralized exchange makes it difficult for a firm's investors to know if the firm has offered patents for sale. Investors are thus unlikely to be able to "see through" the use of patent sales as an earnings management strategy.

Due diligence for a patent purchase can be a lengthy process, at times taking months. The process involves narrowing down the set of patents available for sale based on the purchaser's criteria (Richardson et al. (2019); Kim and Valentine (2023)). However, there are potentially more expedient options for sellers hoping to transact quickly. Between 25 % and 55 % of patent purchases were made by non-practicing entities (NPEs) in the last five years (Richardson et al. (2023)). NPEs may be more likely to serve as liquidity buyers and purchase a group of patents with fewer due diligence requirements in the hope that even one patent will pay off in their litigation strategy.

Patent brokers are another key facilitator of patent sales. Brokers "sell, buy, market, negotiate, evangelize, and promote patents in the market" (Richardson et al. (2023)). If a seller needs to make a patent sale in a timely manner, they could sell patents directly to a broker familiar with the market. Finally, a seller could list a set of patents with a broker well in advance of their fiscal year end to have the option to sell a patent if necessary to meet earnings targets and lower the asking price to expedite the timing of sales as year-end approaches. Discounts on patent asking prices can approach 75 % and tend to increase the longer a patent stays on the market (Richardson et al. (2023)). In sum, sellers in the patent market can use NPEs, patent brokers, or discounts to time their patent transactions.

2.2. Non-practicing entities

Non-practicing entities employ a business model of acquiring patents for the purpose of suing firms ostensibly infringing on those patent rights without engaging in other operating activities such as R&D. Though NPEs are often criticized, they can potentially play a positive role in the economy by providing economies of scale to small companies and inventors without the resources to sue large, well-resourced firms for infringement. Former Federal Trade Commission Chairwoman, Edith Ramirez, has said "the economics of asserting

⁸ In related firm-level consequences tests, we find that firms with a greater proportion of patent sales in month *M* that are core patents have lower future ROA and cash flows. On the other hand, there is no association with future performance when firms are more likely to sell patents that are non-core in month *M*.

patents has traditionally weighed against enforcement by small companies. Large companies may be slow to respond to demand letters, and the cost of a lawsuit may be prohibitive for small patent holders facing an uncertain payoff' (Ramirez (2013)). Furthermore, NPEs can provide liquidity in the secondary patent market, which could increase funding for innovators. Consistent with this notion, Hochberg et al. (2018) find that a liquid patent market can facilitate lending to start-up firms given that such a market provides lenders a liquidation option for patents pledged as collateral.

Corporations lobby politicians for laws that hamper the ability of NPEs to initiate and win patent infringement lawsuits. The Obama administration released a 2013 report on the costs of patent trolls on the economy and innovation. The report notes that in 2011, litigation payments to NPEs totaled \$29 billion, a 400 % increase from 2005, not including any loss in shareholder value. High-tech firms can spend more on patent acquisition and litigation than they do on R&D for new innovations. Gene Sperling, the former director of the National Economic Council, said, "It's clear that the abuse of the patent system is stifling innovation and putting a drag on our economy. The trolling has gotten out of control" (Sperling (2013)). While in aggregate, innovating firms are harmed by costly patent litigation initiated by trolls, the marginal benefit of selling patents to an NPE for a given firm can exceed the marginal cost, especially when the cost of litigation is borne by another party. Thus, selling to trolls to meet earnings targets can produce negative externalities.

2.3. Real earnings management through patent sales

Investors make capital allocation decisions based on comparable and reliable firm performance information. While earnings forecasts can provide an important signal, practitioners and academics alike argue that an excessive focus on meeting earnings targets can induce managers to behave sub optimally (e.g., Dechow and Sloan (1991); Klein (2002); Liu and Ryan (2006); Barton (2011); Barton and Wiseman (2014); Dou et al. (2016); Fu et al. (2020)). This suboptimal behavior can manifest as managers making short-term decisions that boost reported earnings but do not represent the best use of the firm's resources for the long term.

Prior research suggests firms can make suboptimal operating decisions to achieve earnings targets, termed real earnings management. For example, cuts to discretionary expenses, relaxing credit terms, and overproduction of inventory are methods firms can use to boost short-term earnings at the expense of future profitability (Roychowdhury (2006); Cohen et al. (2008); Cohen and Zarowin (2010); Zang (2012); Kothari et al. (2016)).

Managers receive internal information on a firm's performance throughout a fiscal year before financial statements are created and communicated to investors. To execute a real earnings management strategy, a manager must act within the fiscal year (Zang (2012)). Thus, the manager first assesses the firm's expected performance through the end of the fiscal year and engages in real earnings management if it is likely that the firm will miss the target without the costly real action. Accordingly, we expect patent sales motivated by real earnings management to be more likely to take place in the final month of a firm's fiscal year than at other times. In essence, delaying a patent sale until the last month of the year allows managers to have the flexibility and option value to avoid a suboptimal patent sale or preserve their cookie jar reserve for future use if firm performance improves by the end of the year.

Earnings management through patent sales is most closely related to two literatures. First, the literature on real earnings management through asset sales (Bartov (1993)) and second, discretionary cuts to R&D expenditures (Bushee (1998); Kothari et al. (2016); Bereskin et al. (2018)). Bartov (1993) finds that managers strategically time fixed asset sales to manage earnings. He notes that earnings management through asset sales is an attractive tool for managers given that sales timing is in the manager's control and gains are recorded in income from continuing operations that investors utilize in their decision making. However, the precise timing of fixed asset sales is unobservable to the researcher (i.e., only available at the firm-year level), whereas in the patent setting we can observe the date the patent sale was executed. Furthermore, we provide insight into the identity of the counterparty to an earnings management motivated asset sale (i.e., patent trolls), a question previously unexplored in the literature.

When considering the choice of earnings management strategy or mix of strategies to employ, R&D-related strategies are advantageous in that the manager is more informed and has greater expertise than most shareholders. This opacity makes R&D-based earnings management a natural candidate for firms pursuing innovation to consider and related research finds evidence supporting the use of these strategies. Bushee (1998) finds that firms with greater transient institutional investors cut R&D expenditures to avoid earnings declines. Bereskin et al. (2018) find that abnormal cuts to R&D are associated with lower future innovation, as proxied by the number of patents and patent citations. Kothari et al. (2016) find that earnings management via cuts to R&D is associated with lower performance after a seasoned equity offering and argue that R&D cuts are an opaque earnings management channel. As an output of R&D investments, patents and their subsequent sales can be a similarly opaque channel attractive to managers. Taken together, we add to these two streams of research by providing direct evidence on the timing of real earnings management decisions, the choice of assets sold (i.e., internally developed patents with low book values), and the market-wide consequences of selling to certain counterparties (e.g., patent trolls).

Despite the potential costs, Graham et al. (2005) surveyed over four hundred managers and found that executives were more willing to decrease discretionary spending or delay starting a new project to meet earnings targets than changing accounting estimates. A firm's growth critically depends on innovation pursuits. Thus, earnings management tools related to innovation can have a significant impact on a firm's growth potential. While extant research has identified abnormal cuts to R&D as a potential earnings management tool, we expect that firms could also use patent sales to boost earnings to meet or beat earnings targets. Graham et al. (2005) in their survey of executives provide examples of real decisions managers make to achieve financial reporting goals, with one notable example in our context:

“Another example pertains to a company that would sell an internally developed patent to outsiders and recognize revenue or return to meet an earnings target, rather than develop the patent later in-house, even if the expected cash flows associated with in-house development exceed the sale proceeds of the patent.”

This interview provides anecdotal evidence that managers are willing to sell patents to meet or beat earnings expectations. Selling patents the firm could have developed profitably in-house is one form of real earnings management. Another is to strategically time the sale of a “cookie jar reserve” of patents that managers intend to sell anyway to meet earnings targets.

ASC 360–10 Impairment or Disposal of Long-lived Assets requires that firms report gains on patent sales as part of income from continuing operations (see Hanlon et al. (2023) page 13–13). Thus, gains on patent sales will increase an income statement line item to which market participants are particularly attendant. However, these transactions are rarely separately disclosed either in the financial statements, footnotes, or in 8-Ks. This ability to affect income from continuing operations without other salient disclosure makes patent sales a particularly attractive form of earnings management.⁹

An important consideration in understanding strategic patent sales is that managers exercise joint control over both operating decisions and financial reporting choices (Roychowdhury et al. (2019); Huang et al. (2020)). In the presence of moral hazard, managers and shareholders have incentives that are not perfectly aligned, and this joint control creates opportunities for managers to exploit their discretion in mutually reinforcing ways. Specifically, when managers strategically orient their operating decisions (such as selling patents) toward generating short-term income, they also have incentives to alter their reporting and disclosure practices in a manner that reduces transparency about the underlying intent of those actions (Huang et al. (2020)). In the context of patent sales, this joint control manifests as managers’ ability to both time patent transactions strategically and exercise discretion over whether to disclose these transactions through 8-K filings. The discretion involved in determining materiality thresholds for patent sales, combined with the lack of specific disclosure requirements, provides managers with the opportunity to conceal earnings management motivated transactions from investors.

More formally, a firm’s change in patent stock can be derived from the following identity: $\text{initial patent stock} + \text{new grants} - \text{expiries} + \text{patent purchases} - \text{patent sales} = \text{ending patent stock}$. Rearranging terms, $\text{ending patent stock} - \text{initial patent stock} = \text{“change in patent stock”} = \text{new grants} - \text{expiries} + \text{patent purchases} - \text{patent sales}$. That is, a firm’s choice on whether to retain or sell patents (net of $\text{new grants} - \text{expiries} + \text{patent purchases}$) directly impacts a firm’s stock of patents each period. In our context, the firm must determine whether the benefit of meeting earnings targets by selling patents outweighs the cost of selling them. Costs of selling patent rights to manage earnings can include accepting a discounted price to expedite a sale, foregone future licensing revenue, increased frictions in patenting future related inventions, and even failure to develop a profitable project as the Graham et al. (2005) survey anecdote suggests.¹⁰ Thus, as noted above, we argue firms will likely delay the decision to sell patents until the last month of the year to maximize the option value of delaying, given the potentially significant cost of selling patents. These discussions lead to our hypothesis as follows.

Hypothesis. Public firms with incentives to meet or beat earnings expectations sell more patents in the final month of the fiscal year than at other periods during the year.

Nevertheless, there are reasons why we might not observe our predicted outcome. First, if investors can see through the earnings management and discount the value of strategic firms accordingly, firms might not have incentives to manage earnings *ex-ante*. Second, the sale of patents could be prohibitively costly for some firms and may not be a feasible strategy to pursue.¹¹

⁹ As discussed more in Section 4.1.2, 8-K disclosure is not required for material contracts deemed to be made in the ordinary course of business or for contracts on which a firm’s business does not substantially depend. Patent sales, especially sales of non-core patents, are likely to fall into this category of non-disclosure. To better understand the reporting in the 10-K for patent sales, we randomly sampled and inspected one hundred 10-K filings among firm-years in the top quartile of patent sales in month M. We identified only two instances where firms specifically mentioned the transaction of patents in either the footnotes or as a separate line item in the income statement. This procedure corroborates our understanding that patent transactions are unlikely to be easily “undone” by market participants.

¹⁰ The seller can include in the sales agreement a “license back” provision and a covenant not to sue to allow the seller to continue to use the underlying patented technology without fear of infringement litigation. Even though the seller may continue to use the specific patented technology, the seller no longer has the right to enforce that patent in court and obtain licensing fees from other parties. Furthermore, they may still accept a lower price to expedite the strategic timing of the transaction and could face ambiguity in pursuing future innovations in closely related technologies given the often-blurry lines in intellectual property claims. As shown in Table 1 Panel C, relatively few transactions even have an 8-K filing associated with the patent sale and even if there is an 8-K filing with a contract included, portions of the contract can be redacted, making systematic collection of these types of arrangements intractable. Nonetheless, as discussed, even if sale and license back agreements are pervasive, there are still costs to selling patent rights that managers must weigh against the potential benefits.

¹¹ We also recognize the possibility that firms could license their patents to generate income rather than sell them outright if selling patents is too costly. However, whether a firm can recognize revenue at the start of the licensing period depends on the specifics of intellectual property being transferred and the nature of the contract. If the license agreement allows the licensee access to the IP over a period of time and functionality of the IP could significantly change over the period (e.g., through a software update), then revenue will be recognized over time (ASC 606-10-55-62). In those circumstances, managers will unlikely be able to significantly boost *current period* earnings through licensing, making it an unattractive earnings management tool. If managers can successfully structure a licensing contract to allow the recognition of revenue up front, we expect that to bias against us finding results via patent sales.

3. Data and research design

3.1. Data sources and sample selection

We obtain information on patent sales from the USPTO patent assignment data, patent characteristics from PatentsView, firm characteristics from Compustat and CRSP, and analyst forecasts from I/B/E/S. The USPTO patent assignment dataset includes information such as the buyers' and sellers' names, the number of patents included in the sales, the sale execution dates, and a brief description of the reason for the patent trade (see Kim and Valentine (2023); Armstrong et al. (2024)). The PatentsView data allow us to observe patent information such as technology class, the number of figures included in the patent filings, and forward citations received.

We combine the above databases to create a final sample with a firm-month unit of observation. We include firm-months between January 1985 and December 2021 (6,436,358 firm-months). We define the event month as the last month in a firm's fiscal year. We exclude firm months with missing data on total assets, stock prices, and other control variables (leaving 1,633,632 firm-months). Consistent with studies examining real earnings management, we exclude firms in regulated industries (SIC codes between 4400 and 5000), banks, and financial institutions (SIC codes between 6000 and 6500), leaving a sample of 1,333,284 firm-month observations. We then require firms to have at least one patent in their patent stock each month (leaving 1,294,278 the total number of net patent purchases over the past 20 years (i.e., the total number of acquired patents minus the total number of patents sold) updated each month. We use a 20-year period to approximate the length of time a patent right is in force, which is up to 20 years from the patent filing date (35 U.S.C. 154). We apply this requirement because we want to examine the set of firms that have the option to sell a patent each month. Finally, we require non-missing data for control variables, resulting in a final sample of 1,221,216 firm-month observations.

3.2. Variable measurement

3.2.1. Key dependent variable: #patents sold

We define our measure of patent trades, *#Patents Sold*, as the total number of patents sold by a firm in a given month and apply the inverse hyperbolic sine (IHS) transformation. Specifically, we remove a change in patent ownership due to employer assignments, administrative events such as a correction or name change, and security interests where a patent may be pledged as collateral in a loan agreement (Graham et al. (2018)). Thus, *#Patents Sold* identifies a transfer of patent ownership due to trading among entities.

Furthermore, we differentiate between internally developed patents and externally acquired patents, classifying a patent in a transaction as internally generated if the seller is the same entity on the patent grant (i.e., the patent has not been previously sold). Under current GAAP, most expenditures associated with developing patents are expensed, whereas acquired patents are valued on the balance sheet at the purchase price and amortized. As a result, we anticipate that our findings will primarily be concentrated among internally generated patents, which likely allow firms to report larger gains on the income statement due to the asymmetric treatment of acquired and internally developed patents under U.S. GAAP. We also examine whether a patent is core or non-core to a firm's operations. We expect firms to disproportionately sell non-core patents rather than core patents in the final month of the fiscal year, suggesting that firms strategically time the sale of patents they intended to divest, rather than relinquish rights to their core technologies. For presentation purposes, *#Patents Sold* (and other related patent sales measures) are multiplied by 100 unless otherwise noted.

3.2.2. Key independent variables: *M*, *M-1*, *M-2*, ..., *M-5*

To evaluate the timing of the patent sales, we define *M* as an indicator variable that takes on a value of one when it is the last month of a firm's fiscal year. Additionally, we define a set of indicator variables, *M-n* (where *n* ranges from 1 through 5), that take on a value of one when it is *n*th month preceding the final month of the fiscal year. In our sample, 62 % of firms have a December fiscal year end. Consistent with the idea that managers are particularly concerned about fiscal year-end earnings, we expect that strategic patent sales are more likely to occur near the end of the fiscal year in month *M* (Collins et al. (1984); Kerstein and Rai (2007); Das et al. (2009)).¹²

3.3. Research design

To test the hypothesis that firms' strategic patent sales are concentrated at the fiscal year-end, we estimate the following regression:

$$\#Patent\ Sold_{c,t} = \beta M_{c,t} + \beta_1 M - 1_{c,t} + \beta_2 M - 2_{c,t} + \dots + \beta_5 M - 5_{c,t} + \gamma X_{c,t} + \delta_c + \Lambda_t + u_{c,t} \quad (1)$$

The unit of observation is at the firm *c* and month *t*. *#Patents Sold* is the IHS-transformed number of patent sales made by firm *c* in

¹² It is also possible that managers face incentives to manage earnings at the end of each fiscal quarter. To explore this possibility, we create an indicator for whether a month represents a fiscal quarter end (including Q4 at year end) as an alternative independent variable and find that the abnormal patent sales result persists for the final fiscal quarter (untabulated). However, the coefficients on the indicators for the first three fiscal quarters are not significantly different from zero, suggesting that any quarter end incentives to manage earnings are not as powerful as those present at fiscal year end. Also, while our main results using equation (1) only identify the effect for the six months leading up to fiscal year end, our results are not sensitive to including indicators for months *M-10*, *M-9*, ..., *M-1*, *M* and using *M-11* as the comparison month (untabulated).

month t and M , $M-1$, $M-2$, ..., $M-5$ capture the timing of sales relative to the firm's fiscal year end. We cluster standard errors at the firm level. A positive coefficient on β suggests that patent sales are concentrated in the final month of the fiscal year.

X represents a vector of time-varying control variables for firm characteristics that may impact patent market trading unrelated to the timing of sales relative to the fiscal year-end. Specifically, we control for firm characteristics using market capitalization (*Market Cap*) and stock returns (*Monthly Returns*). And we also include the stock of patents (*Patent Stock*) and the number of granted patents (*# Patents Granted*) at month t . We include these controls to account for the possibility that firms with many patents mechanically have more patent sales. Importantly, the stock of patents includes not only the granted or filed patents, but also previously purchased patents. We measure our control variables in month t . Finally, we include fixed effects for the firm-year (δ) and calendar month (Λ) to control for any firm effects specific to a fiscal year, or factors specific to a given month of the year. Taken together, our design exploits *within firm-year* variation in patent sales *timing*, after controlling for any variation in the size of a firm's patent portfolio or the firm's visibility, raising the bar for alternative explanations.

4. Results

4.1. Descriptive statistics

In Fig. 1 Panel A, we present a bar graph of the total number of patent sales by month, compared to the firm's final month in the fiscal year (month M). Across our sample period, sample firms sell in aggregate approximately 20,000 patents per month in the months leading up to fiscal year end. For context, the sample firms had over two million patents granted during the sample period. In the final month of the fiscal year, the number of patents sold jumps to over 27,000 in aggregate. Fig. 1 Panel B scales patent sales by the number of patent grants to adjust for patent portfolio size and the discontinuity at fiscal year-end becomes even more pronounced. These figures provide univariate evidence of increased patent sales in the final month of the fiscal year.

We tabulate descriptive statistics for the firm-months in our sample in Table 1. Panel A presents distributional statistics for the key variables used in our analyses. On average, sample firms sell approximately one-fifth of a patent per month (0.19 after dividing by 100), though the untransformed variable is highly skewed, justifying our use of inverse hyperbolic sine transformed number of patent sales as our primary dependent variable (*#Patents Sold*). This transformation has the advantage of mitigating the influence of outliers while still being defined for firm-months with zero patent sales (Koudijs and Salisbury's (2020); Glaeser and Omartian's (2022)). Approximately 93 % of patents are internally developed, while 7 % are externally acquired (untabulated), which translates into the proportional excess of internally developed patents over acquired patents of 0.83. On average, sample firms have 1.67 patents granted per month and a patent stock of 529 patents, demonstrating that firms in our sample are active innovators that regularly employ patent protection.

Table 1 Panel B presents univariate correlations between *#Patents Sold* and the months leading up to fiscal year end as well as key control variables. There is a significantly positive correlation between *#Patents Sold* and month M , statistically corroborating the trend of higher patent sales in the last month of the fiscal year shown in Fig. 1. Interestingly, there is a significantly *negative* correlation between *#Patents Sold* and month $M-1$, perhaps due to firms holding back on potential patent sales in the period leading up to year end. Intuitively, market capitalization and patenting activity (*Patent Stock* and *# Patents Granted*) are positively linked to *#Patents Sold*. These results provide univariate support for our [hypothesis](#).

4.2. Decision on what patents to sell and disclose

We next provide descriptive statistics on the characteristics of patents sold in the final month of the fiscal year relative to other patents sold in Table 1 Panel C. We find that patents sold in the final month of a firm's fiscal year are of greater value than patents sold in other months of the year (non-month M), using forward citations, patent claims, and patent generality to proxy for patent value. Patents sold in the final fiscal month are also 6 months older, on average, than patents sold in other months of the year, perhaps because more mature patents have a longer proven history that facilitates a timelier sale. As shown in Fig. 2, the ratio of patent value of sold versus retained patents increases as firms approach fiscal year end, suggesting managers sell more valuable patents as the fiscal-period end approaches.¹³ Though descriptive in nature, this suggests that firms are not selling patents of little value to rebalance their innovative portfolio prior to the end of the year.

Despite patents sold in month M being more valuable, they are significantly *less likely* to have an associated 8-K disclosure (18.9 %) relative to patents sold in other months (32.9 %).¹⁴ This pattern is consistent with managers exercising joint control over both operating and disclosure decisions (Roychowdhury et al. (2019); Huang et al. (2020)). When managers strategically time patent sales

¹³ We test this conjecture more formally in an untabulated regression analysis. Specifically, we regress the *Patent Value Index* on M , $M-1$, $M-2$, ..., $M-5$ and our vector of controls and fixed effects. Results indicate that the coefficient on M is significantly positive, confirming that patent value is greater in month M relative to patents sold in other months of the year. *Patent Value Index* is defined as the sum of the quartile rank for each of the following proxies for patent value: *Scaled Citations*, *Number of Claims*, and *Patent Generality* as defined in the Variables Appendix.

¹⁴ To identify patent-related 8-K disclosures, we examine 8-K filings made within five business days of the patent sale execution date that contain the keyword "patent". While the SEC allows firms four business days to file an 8-K after an event, to be conservative, we use five business days to allow for a one-day lag between the firm filing on the EDGAR platform and the 8-K's appearance on the website. For this test as well as the test in Table 2 Column (4) and Table 4 Columns (3) and (4), we limit our sample to 2005–2021 as the list of required 8-K disclosures changed in late 2004.

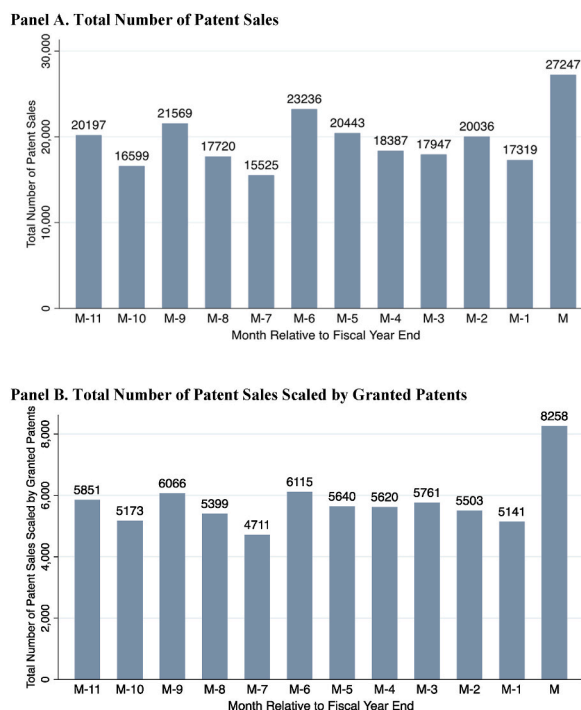


Fig. 1. Number of patent sales by event month.

Notes: Panel A graphs the aggregate number of patent sales by month relative to a firm's fiscal year end from 1985 to 2021. Panel B graphs the aggregate number of patent sales scaled by the firm's patents granted in that month relative to the firm's fiscal year end from 1985 to 2021. Detailed definitions of all variables are provided in [Appendix A](#).

to manage earnings, they simultaneously have incentives to reduce transparency about these transactions to avoid revealing their underlying motivations to investors.

The SEC has required firms to disclose significant events within four business days via an 8-K filing since late 2004 ([Christensen et al. \(2017\)](#); [Noh et al. \(2019\)](#); [He and Plumlee \(2020\)](#)). Patent sales would likely fall under Item 1.01 related to material contracts, which are defined as any contracts "not made in the ordinary course of business" or ordinary business contracts "upon which the registrant's business is substantially dependent" (17 CFR 229.601(b)(10)). The discretion involved in determining the materiality threshold, whether the patent sale is made outside of the ordinary course of business, or if the sale relates to patents on which the firm's business "substantially" depends likely explains why we do not observe an 8-K filing for every patent sale.¹⁵

The significantly lower rate of 8-K filings for month *M* patent sales *relative* to other months of the year could be due to firms selling non-core patents that managers do not consider material. In untabulated tests, we find that the rate of disclosing an 8-K for core patents (34 %) is higher than that of non-core patents (31 %), suggesting that core patents are more likely to be deemed material and thus disclosed. However, within the sale of core patents, we continue to see that month *M* patent sales are disclosed at a lower rate than non-month *M* sales (17 % versus 36 %). Taken together, these descriptive statistics suggest that while patent centrality does indeed play a role in the likelihood of disclosure, it does not likely fully explain the discrepancy in filing rates between month *M* and other months. Thus, this disclosure discrepancy could be at least partially attributable to firms using their available discretion to obfuscate earnings management motivated patent sales. We also examine the disclosure rate of patent sales and the nature of patents sold in a multivariate context as discussed further in Sections 4.2 and 4.3.

For additional context, [Table 1](#) Panel C also includes patent values for the patents that firms chose not to sell (*Unsold Patents*). We find that patents sold in month *M* are relatively *less valuable* than patents that are not sold, suggesting that even strategically motivated

¹⁵ Conceptually, there are competing perspectives on how transaction volume might affect disclosure decisions. Materiality thresholds are legally defined based whether information would "significantly alter the 'total mix' of information made available" to a reasonable investor ([Huang et al., 2024](#); [Thompson et al., 2023](#)), implying that an individual transaction's materiality may not depend on the timing or volume of concurrent transactions. However, one could also envision scenarios where abnormally high volumes of patent transactions in month *M* might practically influence managers' application of materiality judgments through cognitive constraints or perceptions that individual transactions become less important to investors when many similar transactions occur simultaneously. In untabulated tests, we find a negative interaction between month *M* and an indicator for firms in the highest quartile of patents sold in month *M* relative to other months of the year, suggesting that firms with higher relative volumes of month *M* patent sales exhibit stronger reductions in 8-K disclosure rates, providing some empirical support for this latter volume effect.

Table 1
Descriptive statistics.

Panel A. Statistics of Key Variables	Obs.	Mean	SD	P25	Median	P75	
<u>Variables of Interest</u>							
#Patents Sold (untransformed)	1,221,216	19.34	504.92	0.00	0.00	0.00	
#Patents Sold	1,221,216	4.16	33.15	0.00	0.00	0.00	
%Patents Sold with 8K	485,496	0.03	0.37	0.00	0.00	0.00	
Diff in Internally Generated and Externally Acquired Patent Sales	1,221,216	0.83	9.26	0.00	0.00	0.00	
Diff in Non-Core and Core Patent Sales	1,221,216	1.47	10.51	0.00	0.00	0.00	
# Patents Sold to NPE Buyers	1,221,216	0.84	64.13	0.00	0.00	0.00	
Proportion of NPE Buyers	1,221,216	0.88	6.45	0.00	0.00	0.00	
NPE Litigation divided by Total Litigation	1,221,216	0.03	1.71	0.00	0.00	0.00	
Total Net Equity Sales Amount in Q+1	99,526	4.25	10.88	0.00	0.44	4.75	
Total Net # Shares Sold in Q+1	99,526	4.12	10.14	0.00	0.48	4.75	
Total Net # Sales in Q+1	99,526	3.34	5.60	0.00	1.09	4.80	
<u>Control Variables</u>							
Market Cap (untransformed)	1,221,216	3441.09	23651.81	48.43	219.34	1123.50	
Patent Stock (untransformed)	1,221,216	529.43	3907.92	0.00	9.00	71.00	
# Patents Granted (untransformed)	1,221,216	1.67	13.46	0.00	0.00	0.00	
Market Cap	1,221,216	6.20	2.23	4.57	6.08	7.72	
Monthly Returns	1,221,216	0.02	0.59	−0.08	0.00	0.08	
Patent Stock	1,221,216	3.05	2.74	0.00	2.89	4.96	
# Patents Granted	1,221,216	0.32	0.86	0.00	0.00	0.00	
<u>Cross -Sectional Variables</u>							
Meet or Just Beat Firms	1,221,216	0.09	0.28	0.00	0.00	0.00	
Meet or Just Beat Firms (JF)	1,221,216	0.10	0.30	0.00	0.00	0.00	
High Sensitivity to Earnings and Stock Price	1,221,216	0.25	0.43	0.00	0.00	0.00	
Sensitivity to Earnings and Stock Price	1,221,216	−2.91	2.48	−3.78	−3.71	−1.06	
High Financial Constraints Hoberg and Maksimovic (2015)	1,221,216	0.10	0.30	0.00	0.00	0.00	
Financial Constraints Hoberg and Maksimovic (2015)	1,221,216	0.00	0.04	0.00	0.00	0.00	
High Financial Constraints Whited and Wu (2006)	1,221,216	0.21	0.41	0.00	0.00	0.00	
Financial Constraints Whited and Wu (2006)	1,221,216	−0.13	0.23	−0.28	−0.16	0.00	
Panel B. Correlations							
						#Patents Sold	
M						0.0094***	
M-1						−0.0025***	
M-2						−0.0011	
M-3						−0.0003	
M-4						−0.0016*	
M-5						−0.0002	
Market Cap						0.1437***	
Monthly Returns						−0.0006	
Patent Stock						0.2278***	
# Patents Granted						0.2993***	
Panel C. Patent Characteristics by Month of Sale							
	Patents Sold in Month M (N = 52,441)	Patents Sold in Non-Month M (N = 410,849)	Unsold Patents (N = 1,677,508)	Diff. Patents Sold in M vs. Non-M (1)−(2)		Diff. Patents Sold in M vs. Unsold (1)−(3)	
	(1)	(2)	(3)	Diff	t-stat	Diff	t-stat
Forward Citations	14.500	11.879	16.267	2.620***	(13.40)	−1.768***	(-9.28)
Scaled Citations	1.057	0.956	1.089	0.101***	(8.56)	−0.032**	(-2.85)
Number of Claims	15.564	15.103	17.506	0.461***	(7.36)	−1.942***	(-32.30)
Patent Generality	0.316	0.293	0.368	0.023***	(16.44)	−0.053***	(-39.45)
Time to Expiration (months)	155.062	161.471		−6.409***	(-14.72)		
%Patents Sold with 8-Ks	18.96 %	32.91 %		−0.139***	(-58.50)		

Notes: Panel A presents descriptive statistics for the variables used in our analyses, including the mean, standard deviation, and percentile values for our sample of 1,221,216 firm-month observations. We multiply our dependent variables by 100 for presentation purposes. Panel B presents the correlations among #Patents Sold and other variables of interest. Panel C presents the mean and standard deviation for various measures of patent value separately for patents sold in month M and patents sold in other months of the year and presenting the mean of patents filed between 1984 through 2013, which were unsold until 2021. The unit of observation is at the patent level. ***, **, * indicates statistical significance at 1 %, 5 %, and 10 %, respectively (two-tailed). Detailed definitions of all variables are provided in [Appendix A](#).

patent sales are not necessarily the firm's most important patents. Taken together, the descriptive statistics are consistent with firms selling valuable patents in month M to facilitate a sale. However, firms retain patents that are *most* valuable to the firm, likely those central to the businesses' core operations.

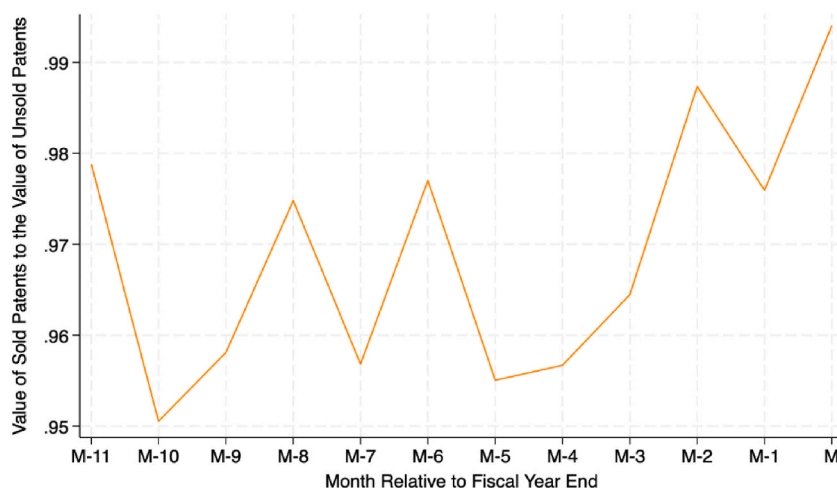


Fig. 2. Average Patent Value of Sold vs. Unsold Patents by Event Month.

Notes: This figure displays the ratio of sold to unsold patents' average "patent value index" relative to firms' fiscal year end. We define patent value index as the sum of the quartile ranks for each of the following proxies of patent value: *Scaled Citations*, *Number of Claims*, and *Patent Generality*, which are defined in the Variables Appendix. We compute the ratio of sold to unsold patents' average patent value index as follows: for each month relative to the fiscal year end, the numerator is the average patent value index across all sold patents in an event month across our sample period. The denominator is the average patent value index across all unsold patents (i.e., patents that were never sold) over our sample period.

4.3. Regression results

Our [hypothesis](#) is that in the presence of earnings pressure firms can sell patents to recognize reported gains and that such earnings-motivated sales are most likely to occur in the final month of the firm's fiscal year. Multivariate regression results are presented in [Table 2](#). Column (1) regresses *#Patents Sold* on *M-5*, *M-4*, ..., *M-1*, *M* without controls and after including firm and year fixed effects. Column (2) incrementally adds time-varying, firm-level control variables as in Equation (1). Column (3) includes the same time-varying controls as in Column (2) but employs firm-year and calendar month fixed effects. We find a significantly positive relation between *M* and *#Patents Sold* in all three columns (coef. = 1.101, t-stat = 8.12 for Column (1); coef. = 1.073, t-stat = 8.03 for Column (2); coef. = 0.715, t-stat = 5.79 for Column (3)). This finding supports our [hypothesis](#) that patent sales are elevated in the final month of a firm's fiscal year. In economic terms, the per month abnormal sales effect ranges from 0.7 % to 1.1 % depending on the specification.¹⁶ This translates into roughly 2600 more aggregate patent sales over our sample period (total average patent sales per year of 6384 (untabulated) x 1.1 % x 37-year sample period = 2598 patents). [Fig. 3](#) graphically shows the regression coefficients from [Table 2](#) Column (3).

Though we do not make predictions for the coefficients on the months leading up to event month *M*, results indicate that firms sell significantly fewer patents in month *M-1* relative to other months of the year (in Columns (1) and (2)) and that the remaining months' sales are generally not statistically distinguishable from the benchmark period. Fewer sales in month *M-1* could be consistent with managers holding back on potential patent sales until month *M*, the firm's final opportunity to recognize a gain prior to fiscal year end. However, we do not observe a statistically significant holding back effect when we employ firm-year and month fixed effects (Column (3)).

We further examine the notion that managers have joint control over both operating and reporting decisions by regressing the probability of 8-K filings on our month indicators ([Roychowdhury et al. \(2019\)](#); [Huang et al. \(2020\)](#)). Column (4) examines the disclosure of these strategic patent sales and shows that the probability of 8-K disclosure is significantly lower in month *M* (coef. = -0.005, t-stat = -2.22), consistent with managers exercising joint control over operating and disclosure decisions to avoid revealing their earnings management motivations.

While the results thus far demonstrate that firms sell more patents in the final month of a firm's fiscal year, the results do not rule out the possibility of other motivations for patent sales that coincide with a firm's fiscal year end. To investigate whether managers sell patents motivated by a desire to achieve earnings targets, we exploit variation in the characteristics of patents sold.

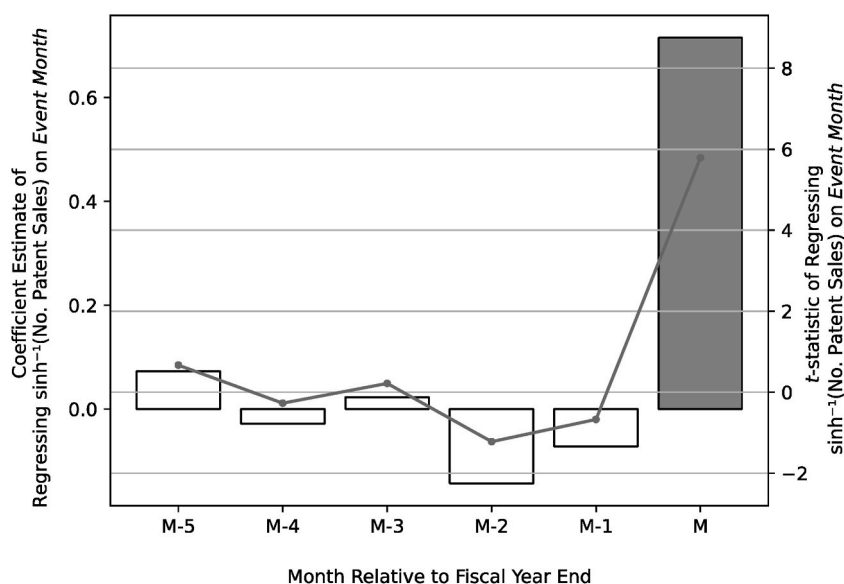
¹⁶ Note that the inverse hyperbolic sine transformation results in coefficients that can be approximately interpreted as percentage changes (see, e. g., and [Koudijs and Salisbury \(2020\)](#) and [Glaeser and Martian \(2022\)](#)). Thus, reported coefficients in [Table 3](#) indicate by what percent *#Patents Sold* are higher in month *M* compared to non-month *M*.

Table 2

#Patents sold over event months.

	#Patents Sold			%Patents Sold with 8K
	(1)	(2)	(3)	(4)
<i>M</i>	1.101*** (8.12)	1.073*** (8.03)	0.715*** (5.79)	-0.005** (-2.22)
<i>M-1</i>	-0.206** (-1.98)	-0.264** (-2.56)	-0.072 (-0.67)	-0.005** (-2.45)
<i>M-2</i>	-0.057 (-0.57)	-0.102 (-1.01)	-0.143 (-1.22)	0.005** (2.08)
<i>M-3</i>	0.034 (0.35)	-0.003 (-0.03)	0.023 (0.22)	-0.010*** (-4.15)
<i>M-4</i>	-0.109 (-1.26)	-0.163* (-1.88)	-0.028 (-0.27)	-0.007*** (-2.87)
<i>M-5</i>	0.048 (0.55)	0.037 (0.43)	0.073 (0.67)	0.004 (1.33)
<i>Market Cap</i>		-0.033 (-0.28)	0.123 (1.31)	0.001 (0.37)
<i>Monthly Returns</i>		0.056* (1.68)	0.036 (1.03)	0.000 (0.77)
<i>Patent Stock</i>		0.967*** (6.80)	10.210*** (18.61)	0.041*** (4.84)
<i># Patents Granted</i>		3.033*** (8.28)	0.268* (1.93)	-0.003 (-1.28)
<i>Obs.</i>	1,221,216	1,221,216	1,221,216	485,496
<i>Adj. R2</i>	0.183	0.185	0.271	0.156
<i>Firm FE</i>	Yes	Yes	No	No
<i>Year FE</i>	Yes	Yes	No	No
<i>Firm-Year FE</i>	No	No	Yes	Yes
<i>Month FE</i>	No	No	Yes	Yes

Notes: This table presents regression results from the regressing #Patents Sold on *M*, *M-1*, *M-2*, ..., *M-5*. Column (1) includes the baseline regression with a vector of control variables and firm and year fixed effects. Column (2) expands the vector of control variables. Column (3) uses firm-year and month fixed effects. The unit of observation is at the firm-month level. Column (4) change the dependent variable as %Patents Sold with 8K. T-statistics, based on standard errors clustered at the firm and year-month level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1 %, 5 %, and 10 %, respectively (two-tailed). Detailed definitions of all variables are provided in [Appendix A](#).

**Fig. 3.** Patent sales relative to public firms' fiscal year end.

Notes: This figure displays the coefficient estimate obtained by regressing #Patents Sold on *M*, *M-1*, *M-2*, *M-3*, *M-4*, *M-5* (represented by the bar graph) along with the corresponding t-statistics (shown in the line graph). The shaded bars signify that the coefficient is significant at the 5 % level. Detailed definitions of all variables are provided in [Appendix A](#).

First, we expect that managers are more likely to sell internally developed patents with a smaller book value than externally acquired patents with a larger book value. The cash a seller receives from a patent sale does not fundamentally depend on whether the invention was developed in-house or externally acquired, and thus if firms sell patents at year end simply for liquidity reasons, we would not expect to observe a difference in the patent's source.¹⁷

In Table 3 Columns (1) and (2), we present the results of regressing *Diff in Internally Generated and Externally Acquired Patent Sales* on $M-5, M-4, \dots, M-1, M$. The dependent variable is calculated as the difference between the number of internally generated patent sales and externally acquired patent sales, scaled by total patent sales (see Appendix A for more detailed variables definitions). We find that both without and with controls (Columns (1) and (2), respectively) there is a significantly positive coefficient on M (coef. = 0.064; t-stat = 3.03 for Column (1); coef. = 0.066; t-stat = 3.14 for Column (2)). This finding is consistent with managers selling disproportionately more internally generated patents in month M than in other months of the year. In fact, the coefficients on $M-5, M-4, \dots, M-1$ suggest managers are no more likely to sell internally generated patents over externally acquired patents in the months leading up to the event month than at other points during the year. In economic terms, internally generated patent sales in month M are 13.9 % higher than internally generated patent sales during other months of the year, relative to baseline averages.¹⁸

Second, we examine whether firms are selling patents in month M that are core to their business versus patents that may result from serendipitous discovery that the firm intends to sell. Patents are unique assets whose value differ depending on the business model of the owner (e.g., an aerospace patent is likely more valuable to Boeing than it would be to IBM). As a result, patents are not as fungible as other assets, such as marketable securities that are more easily replaceable. While anecdotal survey evidence suggests there may be some managers even willing to sell patents that could have been more profitably commercialized to meet an earnings target (Graham et al., 2005), on average, we expect firms to be more willing to sell non-core patents to manage earnings.

In Table 3 Columns (3) and (4), we regress *Diff in Non-Core Patent Sales and Core Patent Sales* on $M-5, M-4, \dots, M-1, M$. The dependent variable is calculated as the difference between the number of non-core patent sales and core patent sales (where core patents are those whose technology area matches the firm's industry), scaled by total patent sales.¹⁹ We find that there is a significantly positive coefficient on M with and without controls (coef. = 0.193; t-stat = 5.20 for Column (3); coef. = 0.198; t-stat = 5.37 for Column (4)). This finding suggests that firms are more likely to strategically time the sale of non-core patents as a means to manage earnings rather than sell patents that are critical to the firm's business activities.²⁰ Taken together, the results in Tables 2 and 3 point towards an earnings management motivation for abnormal patent sales.

4.4. Cross-sectional analyses

We additionally examine two cross-sectional tests to examine whether patent sales in month M are likely to be motivated by earnings management incentives. First, if we observe that final month patent sales are more pronounced in years where a firm either meets or just beats earnings expectations, these patent sales are more likely motivated by earnings pressure. We modify Equation (1) to include an interaction between $M, M-1, M-2, \dots, M-5$ and *Meet or Just Beat* that takes on the value of one when the difference between realized earnings per share and expected earnings per share is between 0 and 1 cent and zero otherwise (see detailed variable definitions in Appendix A). If patent sales in the final month of the fiscal year are indicative of earnings management, we expect the coefficient on $M * \text{Meet or Just Beat}$ to be significantly positive.

Table 4 Columns (1) and (2) examine patent sales as the dependent variable, while Columns (3) and (4) examine the disclosure of

¹⁷ However, we do acknowledge that firms may be less likely to sell a previously purchased patent given that the purchase was made with a purpose, whereas some internally developed patents may be serendipitous discoveries. For this reason, we use the *difference* in internally generated versus externally acquired patent sales as our dependent variable. Absent earnings management motivations, it is more difficult to argue that firms sell *relatively* more internally developed patents precisely at a firm's fiscal year end.

¹⁸ We evaluate the economic effects of our regression coefficient by using the coefficient value (e.g., 0.064 in Column (1) of Table 3) to estimate the implied change in our dependent variable - *Diff in Internally Generated and Externally Acquired Patent Sales* relative to its mean value of the benchmark period $M-6$ through $M-11$ (i.e. 0.46). We derive this change using the equation: $0.064 = 0.46 * (1+r) - 0.46$, where r represents the economic effect of our coefficient relative to the mean value of our dependent variable. By solving for r in this equation, we find that $r = 13.9$ %. This represents the economic magnitude derived from the regression coefficient of 0.064 in Column (1) of Table 3.

¹⁹ We match a patent's CPC tech class with a firm's SIC industry using the cross-referencing tables from Goldschlag et al. (2019). Specifically, we define a patent as a core patent if the firm's industry code corresponds to the top-ranked industry associated with the patent's CPC code. The Goldschlag et al. (2019) mapping links CPC codes to both ISIC and NAICS industry codes. We use the ISIC-based crosswalk because it offers better coverage over our sample period and map each firm's ISIC industry to its historical SIC code in Compustat. Our sample spans from 1985 to 2021. To align with the mapping table, we use ISIC Rev. 2 for years before 1988, ISIC Rev. 3 for 1989–2001, ISIC Rev. 3.1 for 2002–2005, and ISIC Rev. 4 for 2006–2021. Since each CPC code may correspond to multiple ISIC codes, we define a patent as core if the firm's SIC code maps to the industry with the highest probability for that CPC code. Our results remain qualitatively unchanged when expanding the definition to include the top 1–6 industries associated with each CPC code (ranked by probability score).

²⁰ While the results in Table 3 columns 3 and 4 examine within-firm variation in patent sales over the fiscal year, we also consider cross-sectional variation in the sale of core versus non-core patents. Specifically, we measure the HHI of firms' patent filings over the past twenty years (the maximum patent term length) across different technology areas. If firms have a high concentration of patent filings in a certain technology class, they are more likely to have predominately "core" patents. But if a firm is diversified across many technology areas, a firm may be more likely to sell patents to manage earnings as they seem to be filing them for option value. The results of this untabulated analysis show that firms with a low concentration of patent filings are more likely to sell patents in the month M .

Table 3

Patent characteristics and strategic patent sales.

	<i>Diff in</i>		<i>Diff in Non-Core and Core Patent Sales</i>	
	<i>Internally Generated and</i>			
	<i>Externally Acquired Patent Sales</i>			
	(1)	(2)	(3)	(4)
<i>M</i>	0.064*** (3.03)	0.066*** (3.14)	0.193*** (5.20)	0.198*** (5.37)
<i>M-1</i>	-0.025 (-1.22)	-0.019 (-0.92)	-0.043 (-1.21)	-0.028 (-0.80)
<i>M-2</i>	-0.019 (-0.87)	-0.014 (-0.61)	-0.038 (-1.07)	-0.025 (-0.70)
<i>M-3</i>	-0.004 (-0.17)	-0.002 (-0.11)	0.020 (0.59)	0.024 (0.70)
<i>M-4</i>	0.002 (0.11)	0.004 (0.21)	0.005 (0.15)	0.010 (0.32)
<i>M-5</i>	0.000 (0.01)	0.002 (0.11)	0.021 (0.59)	0.026 (0.74)
<i>Market Cap</i>		-0.006 (-0.30)		0.019 (0.62)
<i>Monthly Stock Return</i>		0.008 (1.42)		0.014 (1.36)
<i>Patent Stock</i>		1.515*** (16.72)		2.999*** (18.90)
<i># Patents Granted</i>		0.042 (1.20)		0.069 (1.54)
<i>Obs.</i>	1,221,216	1,221,216	1,221,216	1,221,216
<i>Adj. R2</i>	0.222	0.223	0.273	0.274
<i>Firm-Year FE</i>	Yes	Yes	Yes	Yes
<i>Month FE</i>	Yes	Yes	Yes	Yes

Notes: This table presents regression results from regressing of the *Diff in Internally Generated and Externally Acquired Patent Sales* or *Diff in Non-Core and Core Patent Sales* on *M*, *M-1*, *M-2*, ..., *M-5*. Column (1) includes the baseline regression with a vector of control variables and firm-year and month fixed effects for *Diff in Internally Generated and Externally Acquired Patent Sales*. Column (2) expands the vector of control variables for *Diff in Internally Generated and Externally Acquired Patent Sales*. Column (3) includes the baseline regression with a vector of control variables and firm-year and month fixed effects for *Diff in Non-Core and Core Patent Sales*. Column (4) expands the vector of control variables for *Diff in Non-Core and Core Patent Sales*. The unit of observation is at the firm-month level. T-statistics, based on standard errors clustered at the firm and year-month level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1 %, 5 %, and 10 %, respectively (two-tailed). Detailed definitions of all variables are provided in [Appendix A](#).

these strategic patent sales using *%Patents Sold with 8-K* as the dependent variable. Column (1) shows that the positive link between patent sales and month *M* is amplified for meet or beat firms (coef. = 1.352; t-stat = 3.48). Similarly, Column (3) shows that the reduced disclosure effect is more pronounced for meet or beat firms (coef. = -0.009; t-stat = -2.45), indicating that firms with stronger earnings management incentives are even less likely to disclose their strategic patent sales. In Column (1) and Column (3), we use the most recent analyst consensus forecast for the year, where available, as the market's earnings expectation. However, the most recent analyst consensus forecast for the year could be measured after a firm's fiscal year ends, but before the earnings announcement date.

Real activities management through patent sales requires the manager to finalize a transaction prior to year-end for any associated gain to be recognized in the accounting period. Thus, we also use the market's expected earnings at the time a manager must decide to sell a patent, which we proxy for using the consensus analyst forecast as of the second month of the last fiscal quarter of the year, e.g., November for a December 31st year end firm ([Bhojraj et al. \(2009\)](#)). We present these results in [Table 4](#) Column (2) and Column (4) and find qualitatively similar results (coef. = 1.582; t-stat = 3.75 in Column (2); coef. = -0.010; t-stat = -2.70 in Column (4)).

Taken together, we find that the per month abnormal patent sales effect in month *M* is 1.4 %–1.6 % higher in years where a firm just meets or beats earnings expectations by 1 cent. Moreover, this increase in patent sales for firms that just meet or beat earnings expectations is complemented by the reduced disclosure of patent sales in 8-K filings, providing additional evidence that the reduced transparency is strategically motivated rather than coincidental. This cross-sectional variation further supports the interpretation that managers deliberately reduce disclosure when patent sales are more likely to be driven by earnings management incentives.

[Fig. 4](#) provides graphical evidence corroborating the inferences from [Table 4](#) Columns (1) and (2). Panels A and B are histograms of the average # of *Patents Sold* by *Earnings Forecast Surprise* range (i.e., the difference between actual earnings and market expectations) for patent sales in month *M* and non-month *M*, respectively. Panel A shows that patent sales in the final month of the fiscal year are over three times as common (approximately 0.26/0.07) in years where a firm meets or beats an earnings expectation by one cent than in years where a firm misses expectations by one cent. Importantly, there is no such discontinuity for patent sales made during other months of the year, even when a firm is within one cent of meeting expectations (Panel B). We also present similar graphs using a regression framework that includes our vector of controls as well as firm-year and month fixed effects and do so separately for sales made in month *M* (Panel C) and non-month *M* (Panel D). In this multivariate framework, we observe statistically significant, elevated

Table 4

Patent sales for meet or just beat firm-years.

	#Patents Sold		%Patents Sold with 8K	
	Consensus Forecast	2 Months Prior Consensus Forecast	Consensus Forecast	2 Months Prior Consensus Forecast
	(1)	(2)	(3)	(4)
<i>M * Meet or Just Beat</i>	1.352*** (3.48)	1.582*** (3.75)	-0.009** (-2.45)	-0.010*** (-2.70)
<i>M</i>	0.393*** (5.17)	0.376*** (4.91)	-0.001 (-0.38)	-0.001 (-0.32)
<i>M-1 * Meet or Just Beat</i>	-0.185 (-0.73)	-0.453** (-2.04)	-0.007** (-2.04)	-0.009** (-2.35)
<i>M-2 * Meet or Just Beat</i>	-0.254 (-1.08)	0.124 (0.57)	0.005 (1.24)	0.008* (1.73)
<i>M-3 * Meet or Just Beat</i>	-0.009 (-0.03)	0.054 (0.25)	-0.010*** (-2.83)	-0.011*** (-3.11)
<i>M-4 * Meet or Just Beat</i>	-0.235 (-1.01)	0.130 (0.65)	-0.012*** (-3.35)	-0.010*** (-2.61)
<i>M-5 * Meet or Just Beat</i>	-0.364 (-1.63)	-0.111 (-0.56)	0.008* (1.91)	0.008* (1.79)
<i>M-1</i>	-0.017 (-0.24)	0.015 (0.22)	-0.002 (-0.88)	-0.002 (-0.75)
<i>M-2</i>	-0.006 (-0.08)	-0.045 (-0.59)	0.002 (0.80)	0.001 (0.53)
<i>M-3</i>	0.056 (0.77)	0.046 (0.64)	-0.006** (-2.57)	-0.006** (-2.48)
<i>M-4</i>	0.038 (0.56)	0.005 (0.07)	-0.002 (-0.70)	-0.003 (-1.28)
<i>M-5</i>	0.113 (1.46)	0.090 (1.20)	0.000 (0.12)	0.001 (0.28)
<i>Market Cap</i>	0.041 (0.66)	0.038 (0.60)	0.001 (0.55)	0.001 (0.56)
<i>Monthly Returns</i>	0.027 (1.35)	0.027 (1.34)	0.000 (0.75)	0.000 (0.75)
<i>Patent Stock</i>	6.175*** (19.58)	6.169*** (19.58)	0.041*** (5.62)	0.041*** (5.63)
<i># Patents Granted</i>	0.149 (1.62)	0.138 (1.51)	-0.003 (-1.23)	-0.003 (-1.23)
<i>Obs.</i>	1,221,216	1,221,216	485,496	485,496
<i>Adj. R2</i>	0.276	0.275	0.156	0.156
<i>Firm-Year FE</i>	Yes	Yes	Yes	Yes
<i>Month FE</i>	Yes	Yes	Yes	Yes

Notes: This table presents regression results from regressing #Patents Sold and %Patents Sold with 8K on Meet or Just Beat * M, M-1, M-2, ..., M-5 to examine the moderating effect of firms consistently meeting or beating analyst forecast on the relation between fiscal month and patent sales. We define Meet or Just Beat as an indicator that takes on a value of one when Earnings Forecast Surprise is between 0 and 1 cent. Columns (1) & (3) defines Earnings Forecast Surprise as (actual EPS – analyst forecast EPS). Columns (2) & (4) defines Earnings Forecast Surprise as (actual EPS – analyst forecast EPS 2 months prior to yearend) following the Bhojraj et al. (2009). In both columns, if a firm does not have analyst coverage, we employ prior year's earnings as the benchmark. The unit of observation is at the firm-month level. T-statistics, based on standard errors clustered at the firm and year-month level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1 %, 5 %, and 10 %, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

final month patent sales for periods in which a firm just meets or exceeds earnings targets and observe no effect for patent sales made at other times during the year.

In our second cross-sectional test, we use executives' equity incentives as another proxy for whether managers have strong incentives to meet or beat earnings expectations. Executives believe that issuing earnings targets puts downward pressure on a firm's stock price (Graham et al. (2005)), and thus the more closely tied an executive's compensation is to earnings or stock price, the more likely a manager is to use earnings management techniques such as selling patents for a gain. We regress #Patents Sold on the interaction between *M*, *M-1*, *M-2*, ..., *M-5* and *High Sensitivity to Earnings and Stock Price*. We proxy for high sensitivity to earnings and stock price by first regressing executives cash compensation on changes in ROA and stock returns in each Fama French 12 industry. The sum of the coefficients on changes in ROA and returns measures the direct and indirect effect of earnings' impact on executives' compensation (Leone et al. (2006)). The *High Sensitivity to Earnings and Stock Price* takes on the value of one if a firm's total sensitivity to both earnings and stock prices is in the highest quartile, and zero otherwise. The sample size for these cross-sectional tests is reduced due to the need to have Execucomp database coverage (see Appendix A for variable definitions).

The results are presented in Table 5 Panel A. In Column (1), we find firms whose executives are most sensitive to earnings and stock price sell significantly more patents in month *M* (coef. = 1.087; t-stat = 3.31). In economic terms, *Patents Sold* the per month abnormal patent sales are 1.1 % greater in month *M* for firms when *High Sensitivity to Earnings and Stock Price* is in the highest quartile, relative to firms whose executives are proportionally less sensitive.

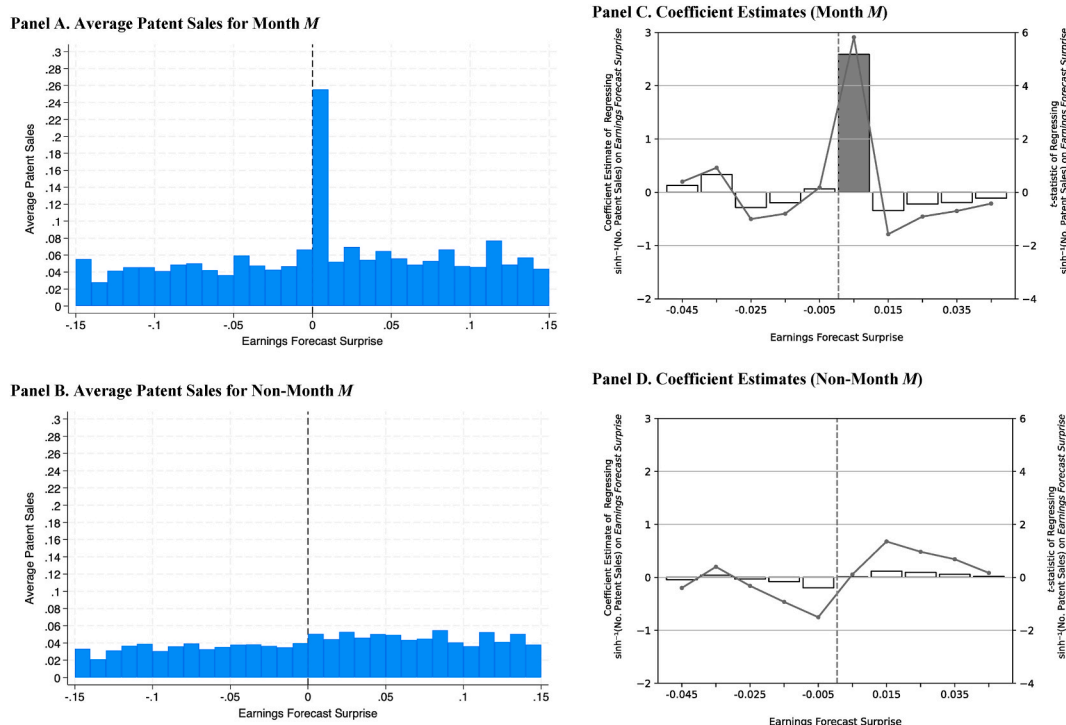


Fig. 4. Earnings forecast surprise and patent sales.

Notes: Panel A and Panel B graphs the average patent sales by earnings forecast surprise range from 1985 to 2021, Panel A shows the average patent sales of month *M* and Panel B shows the average patent sales for other months of the year (Non-Month *M*). Panel C displays the coefficient estimates obtained by regressing #Patents Sold in month *M* on groups of Earnings Forecast Surprise (represented by the bar graph), our vector of controls, and fixed effects along with the corresponding t-statistics (shown in the line graph). Panel D displays the coefficient estimates obtained by regressing #Patents Sold in Non-Month *M* on groups of Earnings Forecast Surprise (represented by the bar graph), our vector of controls, and fixed effects along with the corresponding t-statistics (shown in the line graph). The shaded bars signify that the coefficient is significant at the 5 % level. Detailed definitions of all variables are provided in [Appendix A](#).

We also examine whether executives have more frequent insider equity sales in the quarter following patent sales, indicating these sales are more likely to be motivated by earnings management. Specifically, we regress measures of executives' net equity sales volume on *High Patent Sales in M* and control variables using a firm-year unit of analysis. In [Table 5](#) Panel B, our results indicate that firms with high patent sales in the last fiscal month have 17.5 %–40.5 % more net insider equity sales in the next quarter than firms without high patent sales.²¹ In untabulated analysis, we further decompose these insider sales of stock into routine versus abnormal components following [Cohen et al. \(2012\)](#) and find that the increased insider sales activity is more pronounced in abnormal (opportunistic) sales relative to pre-scheduled transactions. This pattern suggests that managers are capitalizing on the temporary earnings boost from strategic patent sales by opportunistically timing their personal equity liquidations. Taken together, our cross-sectional results point to abnormally high patent sales in the final fiscal month representing strategic patent sales; the firms who sell the greatest number of patents in month *M* are those with high incentives to manage earnings and choose to sell the patents most likely to generate significant reported gains.

4.5. Alternative channels

While cross-sectional results discussed in [Section 4.3](#) demonstrate that final fiscal month patent sales are greater for firms and in time periods with greater incentives to manage earnings, to raise the bar for alternative explanations, we also test three alternative channels. First, prior research indicates that private firms are less likely to have financial pressure distort their operating decisions ([Beatty et al. \(2002\)](#); [Asker et al. \(2015\)](#)). If so, we expect that patent sales in the last month of a firm's fiscal year end are lower for private firms than public firms, which are subject to the scrutiny of the public equity markets. We modify Equation (1) to include the interaction between *M*, *M*-1, *M*-2, ..., *M*-5 and *Private Firm*. *Private Firm* is an indicator variable that takes on the value of one for private firms and zero for publicly traded firms. We identify private firms in the patent assignment data as those that PatentsView identifies as

²¹ For our firm-year level regressions ([Table 5](#) Panel B and [Table 7](#) Panel B and Panel C), we estimate the economic magnitude by dividing the coefficients with the median of the dependent variable.

Table 5
Executives' incentives.

Panel A. Sensitivity to Earnings and Stock Price			
	#Patents Sold		
	(1)	(2)	
<i>M * High Sensitivity to Earnings and Stock Price</i>	1.087*** (3.31)	1.196*** (3.64)	
<i>M</i>	0.428*** (3.15)	0.417*** (3.10)	
<i>M-1 * High Sensitivity to Earnings and Stock Price</i>	-0.120 (-0.43)	-0.041 (-0.15)	
<i>M-2 * High Sensitivity to Earnings and Stock Price</i>	0.390 (1.36)	0.451 (1.57)	
<i>M-3 * High Sensitivity to Earnings and Stock Price</i>	-0.152 (-0.60)	-0.091 (-0.36)	
<i>M-4 * High Sensitivity to Earnings and Stock Price</i>	-0.277 (-1.18)	-0.230 (-0.98)	
<i>M-5 * High Sensitivity to Earnings and Stock Price</i>	0.242 (0.98)	0.264 (1.07)	
<i>M-1</i>	-0.089 (-0.80)	-0.060 (-0.53)	
<i>M-2</i>	-0.282** (-2.40)	-0.252** (-2.12)	
<i>M-3</i>	0.053 (0.43)	0.050 (0.40)	
<i>M-4</i>	0.028 (0.26)	0.033 (0.32)	
<i>M-5</i>	-0.004 (-0.03)	0.009 (0.07)	
<i>Market Cap</i>		0.122 (1.31)	
<i>Monthly Returns</i>		0.036 (1.02)	
<i>Patent Stock</i>		10.222*** (18.65)	
<i># Patents Granted</i>		0.269* (1.94)	
<i>Obs.</i>	1,221,216	1,221,216	
<i>Adj. R2</i>	0.270	0.271	
<i>Firm-Year FE</i>	Yes	Yes	
<i>Month FE</i>	Yes	Yes	
Panel B. Insider Equity Sales			
	Total Net Insider Sales Amount in Q+1 (1)	Total Net # Insider Shares Sold in Q+1 (2)	Total Net # Insider Sales in Q+1 (3)
<i>High Patent Sales in M</i>	0.178*** (5.30)	0.172*** (5.42)	0.191*** (5.57)
<i>Market Cap</i>	0.181*** (14.94)	0.166*** (15.62)	0.180*** (16.49)
<i>Stock Return</i>	-0.001* (-1.88)	-0.001** (-2.12)	-0.002** (-2.32)
<i>Patent Stock</i>	-0.017** (-2.18)	-0.014* (-1.74)	-0.021** (-2.62)
<i># Patents Granted</i>	0.006 (0.76)	0.006 (0.72)	0.008 (0.97)
<i>Obs.</i>	99,526	99,526	99,526
<i>Adj. R2</i>	0.318	0.300	0.392
<i>Year FE</i>	Yes	Yes	Yes
<i>Firm FE</i>	Yes	Yes	Yes

Notes: Panel A presents regression results from regressing *#Patents Sold* on *High Sensitivity to Earnings and Stock Price * M, M-1, M-2, ..., M-5* to examine the moderating effect of high executives' sensitivity to earnings and stock price and patent sales. Column (1) includes the baseline regression with a vector of control variables and firm-year and month fixed effects. Column (2) expands the vector of control variables. The unit of observation is at the firm-month level. T-statistics, based on standard errors clustered at the firm and year-month level, are reported below the coefficient estimates in parenthesis. Panel B presents regression results from the regressing measures of executive equity sales on *High # Patents Sold in M*. The unit of observation is at the firm-year level. T-statistics, based on standard errors clustered at the

firm level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1 %, 5 %, and 10 %, respectively (two-tailed). Detailed definitions of all variables are provided in [Appendix A](#).

a corporation but that are not included as public firms in the updated [Kogan et al. \(2017\)](#) dataset. We use the PatentsView assignee_id variable as a private firm's unique identifier. Because controls such as market capitalization and returns are unavailable for private firms, we omit control variables in this analysis, though we continue to employ firm-year and calendar month fixed effects.²² We further assume all private firms have a December fiscal month end, consistent with the majority of public firms in our sample.

Table 6 Panel A presents the results. We find that private firms are less likely than public firms to sell patents in the last month of the fiscal year, consistent with these firms having less incentive to manage earnings. This is true if we compare the last month of the fiscal year to all other months of the year (Column (1)) or if we map out patent sales in the months leading up to fiscal year end (Column (2)) (i.e., coef. = -1.367; t-stat = -8.73 for Column (1) and coef. = -1.347; t-stat = -8.51 for Column (2)). In terms of economic significance, per month patent sales are 1.4 % lower for private firms in the last month of the fiscal year than public firms.

Importantly, this private firm analysis also provides evidence that tax considerations do not motivate final-month patent sales. Private firms not only face less financial reporting pressure than public firms, but their reporting choices are typically dominated by tax considerations. If tax motivations were driving month *M* patent sales, we would expect private firms to exhibit similar or even stronger patterns given their focus on tax optimization. Instead, the finding that private firms are less likely to sell patents in month *M* suggests that tax reasons are unlikely to explain our main results. In general, firms' tax incentives are to defer gains to future periods, making it unlikely that firms would strategically time patent sales to recognize taxable gains at fiscal year-end.

As a second alternative channel test, we examine whether financially distressed firms exhibit more pronounced patent selling behavior at year end than other firms. We measure financially constrained firms using either the [Hoberg and Maksimovic \(2015\)](#) measure or the [Whited and Wu \(2006\)](#) index and create an indicator variable, *High Financial Constraints* that takes on the value of one if the financial constraint proxy is in the top quartile and zero otherwise. If financially constrained firms sell more patents in month *M* simply for liquidity rather than earnings pressure, we should observe a positive coefficient on *M* * *High Financial Constraints*. The results, presented in **Table 6** Panel B, are inconsistent with this alternative motivation as the interaction term is insignificant (coef. = 0.009; t-stat = 0.03 for Column (1) and coef. = -0.296; t-stat = -1.21 for Column (2)). In sum, all three of the alternative channel tests fail to find evidence that our main results are attributable solely to institutional factors, tax motivation, or financial constraints.

4.6. Other earnings management tools

In untabulated tests, we explore whether managers' use of other earnings management methods affects their propensity to sell patents strategically. We examine interactions between month *M* and indicator variables to identify firms that have already engaged in abnormal R&D cuts in the first three quarters or have low unrealized gains on available-for-sale securities. We find firms are more likely to sell patents in month *M* when they have already used other real earnings management tools, consistent with patent sales serving as a substitute earnings management tool. We also find patent sales increase when accrual-based earnings management is more costly, using Big N auditor presence and shorter operating cycles as proxies for constraints on accruals manipulation. The results show that firms with Big N auditors and shorter operating cycles are more likely to engage in strategic patent sales, consistent with managers substituting toward patent sales as a real earnings management tool when accrual-based methods are costlier to implement ([Cohen et al. \(2008\)](#)).

4.7. Economic consequences

4.7.1. Firm performance

To investigate the implications of firms' abnormal patent sales in month *M*, we examine whether selling patents in month *M* is linked to future performance. As discussed in Section 4.2, firms could choose to sell patents that are core to the firm's business or simply strategically time the sale of patents and instead sell patents that are not core to the business. If firms sell core patents to manage earnings, we expect it is more likely the firm will bear higher costs from those sales and thus it is more likely that core patent sales are negatively correlated with future performance. However, if firms strategically time the sale of non-core patents they intended to sell anyway, we expect it is more likely that the benefits of meeting an earnings target (e.g., [Bartov et al. \(2002\)](#); [Kasznik and McNichols \(2002\)](#); [Chu et al. \(2019\)](#); [Johnson et al. \(2020\)](#)) exceed the costs.

To test these predictions, we modify Equation (1) to a firm-year unit of analysis and employ future ROA or cash flows as dependent variables (either using year *t+1* or the average future performance for years *t+1* to *t+3*). We also replace the month indicator variables with *High Core Patent Wedge in M* (*Low Core Patent Wedge in M*), defined as firm-years in the highest (lowest) quartile of the difference between the percentage of patents sold in month *M* that are core and the exact same percentage computed across all other months of that fiscal year. We are building on the idea that, conditional on patent sales being high in month *M*, if the patents sold are disproportionately of greater (lower) value – proxied by the abnormal amount of core (non-core) patents sold – relative to other months, we expect future performance to decline (improve or remain unchanged) based on the arguments above. Given the change in units, we present descriptive stats at the firm-year level in **Table 7** Panel A. The regression results are in **Table 7** Panels B and C.

²² Note that the inclusion of firm-year fixed effects subsumes the main effect of *Private Firm* in the regression.

Table 6

Tests for alternative channels.

Panel A. Private Firms		
	#Patents Sold	
	(1)	(2)
<i>M * Private Firm</i>	−1.367*** (−8.73)	−1.347*** (−8.51)
<i>M</i>	0.766*** (6.37)	0.746*** (6.18)
<i>M-1 * Private Firm</i>		0.229* (1.91)
<i>M-2 * Private Firm</i>		−0.061 (−0.54)
<i>M-3 * Private Firm</i>		0.017 (0.15)
<i>M-4 * Private Firm</i>		0.086 (0.80)
<i>M-5 * Private Firm</i>		−0.063 (−0.60)
<i>M-1</i>		−0.159 (−1.54)
<i>M-2</i>		−0.197* (−1.74)
<i>M-3</i>		0.060 (0.58)
<i>M-4</i>		−0.086 (−0.86)
<i>M-5</i>		0.038 (0.36)
Obs.	12,879,604	12,879,604
Adj. R2	0.195	0.195
Firm-Year FE	Yes	Yes
Month FE	Yes	Yes
Panel B. Financial Constraints		
Dependent Variable:	#Patents Sold	
Financial Constraint Measure:	Hoberg and Maksimovic (2015)	Whited and Wu (2006)
	(1)	(2)
<i>M * High Financial Constraints</i>	0.009 (0.03)	−0.296 (−1.21)
<i>M</i>	1.072*** (7.87)	1.135*** (7.32)
<i>M-1 * High Financial Constraints</i>	−0.007 (−0.02)	0.007 (0.04)
<i>M-2 * High Financial Constraints</i>	−0.431 (−1.53)	0.218 (1.18)
<i>M-3 * High Financial Constraints</i>	−0.234 (−0.77)	0.217 (1.10)
<i>M-4 * High Financial Constraints</i>	−0.146 (−0.50)	0.470** (2.50)
<i>M-5 * High Financial Constraints</i>	−0.232 (−0.82)	−0.187 (−1.03)
<i>M-1</i>	−0.264** (−2.52)	−0.266** (−2.20)
<i>M-2</i>	−0.059 (−0.58)	−0.148 (−1.25)
<i>M-3</i>	0.020 (0.20)	−0.049 (−0.43)
<i>M-4</i>	−0.149 (−1.55)	−0.263** (−2.43)
<i>M-5</i>	0.060 (0.60)	0.076 (0.68)
<i>Market Cap</i>	−0.036 (−0.31)	−0.035 (−0.29)
<i>Monthly Returns</i>	0.058* (1.77)	0.058* (1.76)
<i>Patent Stock</i>	0.966*** (6.99)	0.966*** (6.98)

(continued on next page)

Table 6 (continued)

Panel B. Financial Constraints		
Dependent Variable:	#Patents Sold	
Financial Constraint Measure:	Hoberg and Maksimovic (2015)	Whited and Wu (2006)
	(1)	(2)
# Patents Granted	3.044*** (8.54)	3.044*** (8.55)
Obs.	1,221,216	1,221,216
Adj. R2	0.185	0.185
Firm-Year FE	Yes	Yes
Month FE	Yes	Yes

Notes: Panel A presents regression results from regressing #Patents Sold on Private Firm * M, M-1, M-2, ..., M-5 to examine the moderating effect of private firm preceding the fiscal year end and patent sales. We define Private Firm as a dummy variable that takes on a value of one when the firm is a non-public firm and zero otherwise. In Panels A, columns (1) include the baseline regression. Columns (2) augment the vector of control variables. In Panel B, we define High Financial Constraints as a dummy variable that takes on a value of one when the firm-year value of either the Hoberg and Maksimovic (2015) measure (column 1) or the Whited and Wu (2006) measure (column 2) is in the highest quartile and zero otherwise. The unit of observation is at the firm-month level. T-statistics, based on standard errors clustered at the firm and year-month level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1 %, 5 %, and 10 %, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

We find firms with a greater wedge of patent sales in month *M* that are core patents have lower future ROA and cash flows. Specifically, Panel B Columns (1) and (2) show a negative link between High Core Patent Wedge in *M* and future ROA ((coef. = -0.018; t-stat = -2.29 for Column (1) and coef. = -0.012; t-stat = -3.10 for Column (2)).²³ Columns (3) and (4) show a negative link between High Patent Sales in *M* and future cash flows (coef. = -0.016; t-stat = -2.96 for Column (3) and coef. = -0.011; t-stat = -3.52 for Column (4)). These estimates translate into a -10.9 % to -16.4 % effect on ROA, and a -13.8 % to -22.9 % effect on cash flows for firms with a greater proportion of core patent sales in the last month of the fiscal year, compared to their respective sample median values. These findings are consistent with the interpretation that firms that sell core patents to manage earnings experience future performance declines.²⁴ However, note that our tests of future performance are not necessarily causal. It remains possible that managers increase the sales of core patents when they expect future earnings to decline.

In Table 7 Panel C, we examine firms in the bottom quartile of the difference between the percentage of patents sold in month *M* that are core and the exact same percentage computed across all other months of that fiscal year (i.e., Low Core Patent Wedge) and find minimal association with future performance. Specifically, Panel C Columns (1) and (2) show no statistically significant association between Low Patent Value Wedge in *M* and future ROA, and the magnitude of the coefficients is much smaller than the coefficients for firms in the top quartile (coef. = -0.005; t-stat = -0.89 for Column (1); coef. = -0.005; t-stat = -1.55 for Column (2)). Columns (3) and (4) show a small negative association with future cash flows (coef. = -0.005; t-stat = -1.27 for Column (3); coef. = -0.003; t-stat = -1.07 for Column (4)), but the economic magnitude is substantially smaller than the effects observed for firms with high core patent wedges. This suggests that the negative future performance consequences from strategic patent sales manifest primarily among firms that disproportionately sell core patents in month *M*, while strategically timing the sale of non-core patents (reflected in a low wedge) is significantly less costly to firms. The contrast between high and low wedge results supports the interpretation that managers can minimize the costs of strategic patent sales by focusing on non-core patents they intended to divest anyway, rather than being forced to sell valuable core technologies to meet earnings targets.²⁵

²³ Gains or losses on the sale of finite lived intangible assets are reported on the income statement as a component of income from continuing operations (Hanlon et al. (2023) page 13-13). If a firm sells patents in year *t* to manage earnings upward, then it is possible that earnings from continuing operations is lower in year *t+1* and years *t+1* to *t+3* not due to a decline in future performance but simply because of reversion to the mean. To avoid this alternative interpretation, we use operating income before depreciation (i.e., a subtotal of income that does not include gains on patent sales) as our proxy for performance. Also, we also employ cash flow from operations as a proxy for future performance, which does not include cash flows from patent sales, and find consistent results. While prior research has shown that investors weight income statement line items differently (McVay (2006); Bradshaw and Sloan (2002)), the extant literature also confirms that investors fixate on bottom-line earnings (i.e., income before extraordinary items) (Sloan (1996); Bradshaw et al. (2001); Richardson et al. (2005); Shi and Zhang (2012)). Thus, managers still have an incentive to manage earnings via patent sales whose gains are included in income from continuing operations.

²⁴ In untabulated analysis, we regress future R&D and patent sales in year *t+1* or years *t+1* to *t+3* on High Patent Sales in *M* that is an indicator taking on the value of one if patent sales were in the top quartile and zero otherwise. We find a 0.8 %–1.3 % increase in future R&D expenditures and a 7.0 %–10.2 % decrease in future patent sales for firms with high patent sales. This pattern of results is consistent with firms increasing their investment in R&D and reducing future patent sales, perhaps in an effort to recover the potential value lost.

²⁵ In untabulated analyses, we define High (Low) Core Patent Wedge in *M* by comparing the average quality of patents sold in month *M* to that of in other months. Using citations-based measures to proxy a patent's quality, we find similar results where High (Low) Core Patent Wedge in *M* is linked to low (high) future performance.

Table 7
Firm consequences.

Panel A. Future Performance Consequences, Descriptives						
	Obs.	Mean	SD	P25	Median	P75
<u>Dependent Variables</u>						
ROA (t+1)	93,081	0.07	1.55	0.02	0.11	0.18
ROA (avg, t+1~t+3)	78,732	0.08	1.55	0.04	0.11	0.18
Cash Flow/Asset (t+1)	87,896	0.02	0.37	-0.00	0.07	0.14
Cash Flow/Asset (avg, t+1~t+3)	74,086	0.03	0.29	0.02	0.08	0.13
<u>Independent Variables</u>						
Core Patent Wedge	93,081	-0.007	0.216	0.00	0.00	0.00
High Core Patent Wedge	93,081	0.002	0.049	0.00	0.00	0.00
Low Core Patent Wedge	93,081	0.009	0.092	0.00	0.00	0.00
<u>Control Variables</u>						
Market Cap	93,081	6.22	2.22	1.69	4.60	6.10
Stock Return	93,081	0.24	4.68	-0.84	-0.27	0.00
Patent Stock	93,081	3.09	2.74	0.00	0.00	3.00
# Patents Granted	93,081	1.09	1.67	0.00	0.00	0.00

Panel B. Future Performance Consequences for High Core Patent Wedge, Regressions				
	ROA (t+1)	ROA (avg, t+1~t+3)	Cash Flow/Asset (t+1)	Cash Flow/Asset (avg, t+1~t+3)
	(1)	(2)	(3)	(4)
High Core Patent Wedge in M	-0.018** (-2.29)	-0.012*** (-3.10)	-0.016*** (-2.96)	-0.011*** (-3.52)
Market Cap	0.014** (2.04)	0.001 (0.47)	0.007*** (3.71)	0.004*** (2.78)
Stock Return	0.000 (1.14)	0.000 (1.02)	0.000 (1.15)	0.000 (1.06)
Patent Stock	-0.021 (-1.22)	-0.001 (-1.01)	0.003 (1.46)	0.001 (1.01)
# Patents Granted	-0.001 (-0.34)	-0.004*** (-3.64)	-0.002* (-1.82)	-0.003*** (-3.05)
ROA (t)	0.336*** (4.54)	0.132*** (9.33)	0.206*** (8.73)	0.116*** (11.06)
Cash Flow/Asset (t)	0.254*** (3.72)	0.162*** (9.53)	0.227*** (6.81)	0.120*** (8.63)
Obs.	93,081	78,732	87,896	74,086
Adj. R2	0.255	0.798	0.386	0.775
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Panel C. Future Performance Consequences for Low Core Patent Wedge, Regressions				
	ROA (t+1)	ROA (avg, t+1~t+3)	Cash Flow/Asset (t+1)	Cash Flow/Asset (avg, t+1~t+3)
	(1)	(2)	(3)	(4)
Low Core Patent Wedge in M	-0.005 (-0.89)	-0.005 (-1.55)	-0.005 (-1.27)	-0.003 (-1.07)
Market Cap	0.014** (2.04)	0.001 (0.47)	0.007*** (3.70)	0.004*** (2.78)
Stock Return	0.000 (1.14)	0.000 (1.02)	0.000 (1.15)	0.000 (1.06)
Patent Stock	-0.021 (-1.22)	-0.001 (-1.00)	0.003 (1.46)	0.001 (1.01)
# Patents Granted	-0.001 (-0.35)	-0.004*** (-3.64)	-0.002* (-1.82)	-0.003*** (-3.05)
ROA (t)	0.336*** (4.54)	0.132*** (9.32)	0.206*** (8.73)	0.116*** (11.06)
Cash Flow/Asset (t)	0.254*** (3.72)	0.162*** (9.53)	0.227*** (6.81)	0.120*** (8.62)
Obs.	93,081	78,732	87,896	74,086
Adj. R2	0.255	0.798	0.386	0.775
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Notes: Panel A presents descriptive statistics at the firm-year level for the regression variables used in Table 7 Panel B and Panel C. Panel B presents regression results from the regressing measures of future performance on *High Core Patent Wedge in M* while panel C presents regression results from regressing future performance on *Low Core Patent Wedge in M*. The unit of observation is at the firm-year level. T-statistics, based on standard errors clustered at the firm level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1 %, 5 %, and 10 %, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

4.7.2. Implications for the market for innovation: patent trolls as strategic liquidity providers

Next, we examine the ways in which patents sold in month M impact the market for innovation more broadly. We find that NPEs play an important role in facilitating patent transactions specifically in month M . In our sample, 55 % of sales made to NPEs occur in the last fiscal month (untabulated). Motivated by this observation, we examine NPEs as purchasers of month M patents and the attendant implications for the litigation environment. We begin by regressing *# Patents Sold to NPE Buyers* on M , $M-1$, $M-2$, ..., $M-5$ and controls and present the results in Table 8 Panel A. We find the number of patents sold to NPE buyers is greater when the patent is sold in month M (Column (1) (coef. = 0.045; t-stat = 2.81)). However, given that we find a greater number of patents sold in month M in general, it is perhaps unsurprising that the number of NPE buyers increases.

Accordingly, we next investigate whether NPEs represent a greater *proportion* of month M buyers. In Column (2) of Table 8 Panel A, we find a significantly positive association between patents sold in month M and *Proportion of NPE Buyers* (coef. = 0.070; t-stat = 3.12), suggesting that a patent sold in the last month of the fiscal year is more likely to be purchased by an NPE than if the patent is sold at another point during the year. Finally, we examine whether NPEs are involved in subsequent litigation involving month M patent sales. In Column (3) of Table 8 Panel A we find that the proportion of litigation cases involving NPEs (defined as the number of sold patents litigated by NPE asserters divided by the total number of sold patents involved in litigation and set to zero for firm-months with no litigation events) increases significantly when the patents were sold in the last month of a firm's fiscal year end.²⁶

However, it is possible that because NPEs are a liquidity provider in the secondary patent market that their bargaining power increases in the last month of the fiscal year when firms attempt to sell patents with less notice when motivated by earnings management. The observation that patents sold in month M are more valuable than those sold in other months is consistent with an increase in NPE bargaining power. If NPEs have more bargaining power or increase their patent purchases for other reasons coincident with firms' fiscal year ends, the patent sales we observe could be attributable to NPE purchasing behavior rather than earnings management incentives.²⁷ Accordingly, we examine whether the number of patents sold to NPEs and litigation outcomes are incrementally greater when firms' incentives to manage earnings are higher. We interact M with *Meet or Just Beat* and reperform the analyses in Table 8 Panel A. Table 8 Panel B shows that the increased selling to NPEs and related NPE-specific litigation is concentrated among firms that just meet or beat earnings expectations (Columns (1) through (3)). This is consistent with earnings management motivations contributing to month M patent sales to NPEs.

4.8. Additional analyses

4.8.1. Robustness

Finally, we examine the robustness of our main result to various measurement alternatives and present the results in Table 9. First, we exclude firms in our sample that have December fiscal year ends (approximately 60 % of sample firms) and find our results continue to hold (Column (1)). This suggests that there is not a specific institutional feature of the patent market related solely to the month of December that would explain our results. Second, our inferences remain unchanged if we alter the fixed effect structure to employ industry by year fixed effects (Column (2)); or firm and industry by year fixed effects (Column (3)). Third, we scale the dependent variable, *Patents Sold*, by total assets or patent stock in Columns (4) and (5), respectively and find consistent results. Finally, we use a Poisson regression in Column (6) and find our inferences are unchanged.

5. Conclusion

In sum, we find an abnormal level of patent sales in the last month of public firms' fiscal years. These patent sales appear to be a result of firms engaging in earnings management rather than institutional features of the patent market or for tax reasons. Taken together, our results are most consistent with the interpretation that capital market pressure to meet earnings targets can result in firms using patent sales to boost reported earnings. Our findings indicate that reporting rules for internally developed intangible assets such as patents could have the unintended consequence of making patent sales more attractive to firms under earnings pressure.

Importantly, patents sold in the final month of the fiscal year are more likely to be litigated in general, to be sold to "patent trolls", and are more likely to be involved in subsequent patent troll litigation, suggesting that strategic patent sales can have socially costly consequences. These results should be of interest to academics interested in patent assignments and litigation, to regulators considering optimal disclosure regulation around intangible assets, and to policymakers examining short-term incentives in public markets.

²⁶ Our dependent variable measures the proportion of NPE litigated patents relative to total litigated patents for each firm-month. Note that the observation count in Column (3) of Table 8 is the same as the first two columns (1,221,216) because we assume a value of zero if a firm-month has no litigation event. An alternative approach is to limit our sample to firm-months with at least one litigated patent (reducing observations to 149,412) and use the number of patents litigated by NPEs as the dependent variable. This approach yields qualitatively similar inferences regarding the relationship between month M patent sales and NPE litigation patterns, confirming the robustness of our findings (untabulated).

²⁷ This increased bargaining power most likely manifests when a larger number of firms' sales cluster in time, such as firms with a 12/31 fiscal year end. In Table 9 we show our results obtain even among the subset of firms with non-December fiscal year ends, suggesting that NPE purchasing behavior is not the sole factor explaining our results.

Table 8

Characteristics and subsequent events for patents sold in month M.

Panel A. Baseline Specification			
	<u># Patents Sold</u> <u>to NPE Buyers</u>	<u>Proportion of</u> <u>NPE Buyers</u>	<u>NPE Litigation</u> <u>divided by Total Litigation</u>
	(1)	(2)	(3)
<i>M</i>	0.045** (2.81)	0.070*** (3.12)	0.023*** (2.75)
<i>M-1</i>	-0.027 (-1.75)	-0.020 (-0.93)	0.003 (0.44)
<i>M-2</i>	0.009 (0.39)	0.001 (0.06)	0.001 (0.15)
<i>M-3</i>	0.018 (1.22)	0.014 (0.64)	0.003 (0.58)
<i>M-4</i>	-0.014 (-0.86)	0.005 (0.22)	-0.007 (-1.17)
<i>M-5</i>	0.022 (1.20)	0.017 (0.74)	-0.002 (-0.23)
<i>Market Cap</i>	0.018** (2.91)	-0.001 (-0.07)	-0.005 (-0.84)
<i>Monthly Returns</i>	0.001 (0.33)	0.006 (1.39)	-0.000 (-0.46)
<i>Patent Stock</i>	0.127*** (9.94)	1.200*** (16.55)	0.089*** (3.43)
<i># Patents Granted</i>	0.307*** (8.75)	0.027 (0.90)	0.000 (0.06)
<i>Obs.</i>	1,221,216	1,221,216	1,221,216
<i>Adj. R2</i>	0.060	0.220	0.050
<i>Firm-Year FE</i>	Yes	Yes	Yes
<i>Month FE</i>	Yes	Yes	Yes
Panel B. Moderating Effect of <i>Meet or Just Beat</i>			
	<u># Patents Sold</u> <u>to NPE Buyers</u>	<u>Proportion of</u> <u>NPE Buyers</u>	<u>NPE Litigation</u> <u>divided by Total Litigation</u>
	(1)	(2)	(3)
<i>M * Meet or Just Beat</i>	0.006*** (2.60)	0.003** (1.98)	0.050** (2.05)
<i>M</i>	-0.001 (-0.96)	-0.001 (-1.10)	0.012 (1.48)
<i>M-1</i>	-0.001 (-1.32)	-0.001 (-1.52)	0.003 (0.54)
<i>M-2</i>	-0.000 (-0.20)	-0.000 (-0.35)	0.002 (0.44)
<i>M-3</i>	-0.000 (-0.59)	-0.000 (-0.77)	-0.002 (-0.29)
<i>M-4</i>	0.000 (0.00)	-0.000 (-0.33)	-0.013** (-2.10)
<i>M-5</i>	-0.000 (-0.08)	-0.000 (-0.30)	0.002 (0.25)
<i>Market Cap</i>	-0.000 (-0.61)	-0.000 (-0.93)	-0.005 (-0.92)
<i>Monthly Returns</i>	0.000 (0.04)	-0.000 (-0.02)	-0.000 (-0.44)
<i>Patent Stock</i>	0.009*** (4.96)	0.006*** (4.85)	0.090*** (3.46)
<i># Patents Granted</i>	-0.000 (-0.28)	-0.000 (-0.12)	0.000 (0.06)
<i>Obs.</i>	1,221,216	1,221,216	1,221,216
<i>Adj. R2</i>	0.184	0.177	0.050
<i>Interactions with M-X</i>	Yes	Yes	Yes
<i>Firm-Year FE</i>	Yes	Yes	Yes
<i>Month FE</i>	Yes	Yes	Yes

Notes: Panel A of this table presents regression results from regressing the # *Patents Sold to NPE Buyers* on *M*, *M-1*, *M-2*, ..., *M-5* in Column (1), regressing the *Proportion of NPE Buyers* on *M*, *M-1*, *M-2*, ..., *M-5* in Column (2), and regressing the *NPE Litigation divided by Total Litigation* on *M*, *M-1*, *M-2*, ..., *M-5* in Column (3). Panel B utilizes the same dependent variables as in Panel A but considers the interaction of *M * Meet or Just Beat*. All results include the baseline regression with a vector of control variables and firm-year and month fixed effects. The unit of observation is at the firm-month level. T-statistics, based on standard errors clustered at the firm and year-month level, are reported below the coefficient estimates in parenthesis. ***,

***, * indicates statistical significance at 1 %, 5 %, and 10 %, respectively (two-tailed). Detailed definitions of all variables are provided in [Appendix A](#).

Table 9
Robustness.

	#Patents Sold	#Patents Sold	#Patents Sold	#Patents Sold (scaled by Assets)	#Patents Sold (scaled by Patent Stock)	Untransformed #Patents Sold (Poisson)
	(1)	(2)	(3)	(4)	(5)	(6)
<i>M</i>	0.551*** (3.88)	1.090*** (8.09)	1.079*** (8.01)	0.015** (2.18)	0.045*** (3.92)	0.131*** (4.89)
<i>M-1</i>	-0.015 (-0.12)	-0.314*** (-3.16)	-0.253** (-2.48)	0.006 (0.85)	0.002 (0.30)	0.011 (0.36)
<i>M-2</i>	0.026 (0.19)	-0.137 (-1.39)	-0.093 (-0.93)	0.000 (0.02)	0.016** (2.05)	-0.018 (-0.59)
<i>M-3</i>	0.131 (0.93)	-0.027 (-0.30)	0.006 (0.06)	0.009 (1.27)	0.012 (1.50)	0.015 (0.56)
<i>M-4</i>	-0.113 (-0.91)	-0.236*** (-2.76)	-0.151* (-1.78)	0.003 (0.57)	0.006 (0.86)	0.002 (0.07)
<i>M-5</i>	0.128 (0.84)	0.036 (0.43)	0.041 (0.48)	-0.015*** (-2.68)	-0.004 (-0.41)	0.016 (0.53)
<i>Market Cap</i>	0.128 (0.91)	0.229** (2.20)	-0.084 (-0.75)	-0.005 (-0.49)	-0.007 (-0.74)	-0.009 (-0.29)
<i>Monthly Returns</i>	0.075 (1.48)	0.057* (1.69)	0.060* (1.79)	-0.004 (-0.51)	-0.001 (-0.59)	0.016 (1.16)
<i>Patent Stock</i>	9.461*** (13.62)	0.926*** (14.41)	0.727*** (3.55)	0.653*** (12.07)	1.962*** (13.34)	3.356*** (13.35)
<i># Patents Granted</i>	0.461** (1.97)	9.041*** (10.86)	2.465*** (8.42)	0.003 (0.51)	0.019*** (3.67)	0.011 (0.75)
<i>Obs.</i>	464,580	1,221,216	1,221,216	1,221,216	1,120,089	1,221,216
<i>Adj./Pseudo R2</i>	0.293	0.119	0.198	0.019	0.026	0.566
<i>Cluster</i>	Firm	Firm	Firm	Firm	Firm	Firm
<i>Sample</i>	No Dec. Fis. End	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample
<i>Firm FE</i>	No	No	Yes	No	No	No
<i>Firm-Year FE</i>	Yes	No	No	Yes	Yes	Yes
<i>Month FE</i>	Yes	No	No	Yes	Yes	Yes
<i>Industry by Year FE</i>	No	Yes	Yes	No	No	No

Notes: This table presents robustness tests using alternative sample composition, fixed effect structures, or dependent variable measurement. The unit of observation is at the firm-month level. T-statistics, based on standard errors clustered at the firm and year-month level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1 %, 5 %, and 10 %, respectively (two-tailed). Detailed definitions of all variables are provided in [Appendix A](#).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Variables Definitions

Variable	Definition	Source
Dependent Variables		
<i>#Patents Sold</i>	The IHS-transformed number of patents sold in a firm-month. The value is multiplied by 100 for presentation purposes.	Patent Assignment Data
<i>%Patents Sold with 8K</i>	The IHS-transformed number of patents sold in a firm-month with 8-K divided by the number of patents sold with or without 8-K. If a firm-month has no patent sold, the variable takes on the value of zero. The value is multiplied by 100 for presentation purposes.	Patent Assignment Data, SEC Edgar
<i>Diff in Internally Generated and Externally Acquired Patent Sales</i>	$(\# \text{ Internally Generated Patent Sales} - \# \text{ Externally Acquired Patent Sales}) / (1 + \# \text{ Internally Generated Patent Sales} + \# \text{ Externally Acquired Patent Sales})$	Patent Assignment Data
<i># Internally Generated Patent Sales</i>	Number of internally generated patents sold in a firm-month. For a patent to be considered internally generated in a transaction, we verify that all sellers in this transaction have not previously been one of the buyers in any past transactions involving this patent.	Patent Assignment Data

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Variable	Definition	Source
# Externally Acquired Patent Sales	Number of externally acquired patents sold in a firm-month. For a patent to be considered externally acquired in a transaction, we verify that one of the sellers in this transaction was one of the buyers in the most recent sale of the patent.	Patent Assignment Data
Diff in Non-Core and Core Patent Sales	$(\# \text{ of Non-Core Patent Sales} - \# \text{ Core Patent Sales}) / (1 + \# \text{ of Non-Core Patent Sales} + \# \text{ Core Patent Sales})$	PatentsView, Compustat
# of Core Patent Sales	The number of core patents sold. Core patents are defined as patents whose technology class matches the primary 3-digit SIC code of the firm using the patent technology class to industry classification crosswalk developed in Goldschlag et al., (2019) . A patent is defined as a core patent if the firm's industry code corresponds to the top-ranked industry associated with the patent's CPC code.	PatentsView, Compustat
# of Non-Core Patent Sales	The number of patents sold that do not meet the definition of a core patent.	PatentsView, Compustat
Total Net Insider Sales Amount in Q+1	The IHS-transformed executives' net equity sales amount (sales amount minus purchase amount) in Q+1 scaled by last year's average net sales amount.	LSGE Insiders Data
Total Net # Insider Shares Sold in Q+1	The IHS-transformed net number of executives' shares sold (number of shares that are sold minus number of shares that are purchased) in Q+1 scaled by last year's average net number of shares sold.	LSGE Insiders Data
Total Net # Insider Sales in Q+1	The IHS-transformed number of executives' net equity sales transactions (number of sales transactions minus number of purchase transactions) in Q+1 scaled by last year's average net number of equity sales transactions.	LSGE Insiders Data
# Patents Sold to NPE Buyers	The IHS-transformed number of patents sold to NPE buyers.	Patent Assignment Data, NPE Litigation Dataset
Proportion of NPE Buyers	The IHS-transformed number of patents sold to NPE buyers divided by the number of patents sold. If a firm-month has no patents sold to NPE buyers, the variable takes on the value of zero.	Patent Assignment Data, NPE Litigation Dataset
NPE Litigation divided by Total Litigation	The IHS-transformed number of sold patents that is litigated by NPE asserters divided by the number of sold patents that is litigated. If a firm-month has no litigation events, the variable takes on the value of zero.	Patent Assignment Data, NPE Litigation Dataset
#Patents Sold (scaled by Assets)	The IHS-transformed number of patents sold divided by total assets in a firm-month.	Patent Assignment Data, Compustat
#Patents Sold (scaled by patent stock)	The IHS-transformed number of patents sold divided by patent stock of the previous fiscal year-end month.	Patent Assignment Data, Patents View, Kogan et al. (2017)
Forward Citations	The number of forward citations a patent receives.	Patent Assignment Data, PatentsView
Scaled Citations	<i>Forward Citations</i> scaled by the average number of forward citations received by all patents granted in the same tech class-year.	Patent Assignment Data, PatentsView
Number of Claims	The number of claims in the patent.	Patent Assignment Data, PatentsView
Patent Generality	The concentration of forward citations received across technology classes, calculated as $Generality_i = 1 - \sum_j s_{ij}^2$, where s_{ij} represents the fraction of citations patent i receives from patent class j out of n_i patent classes.	Patent Assignment Data, PatentsView
Time to Expiration	The number of months between the date a patent is sold and its expiration date. Before 1995, patents generally had a life of 17 years from the date of issuance; after 1995, the patent life changed to 20 years from the date of application.	Patent Assignment Data, PatentsView
Patent Age at Time of Sale	The number of months between when a patent is filed and when it is sold.	Patent Assignment Data, PatentsView
%Patents Sold with 8-Ks (patent level)	The percentage of patent sales that have at least one 8K disclosure containing the keywords "patent," "patents," or "patenting" within 5 business days after the patent transaction date.	Patent Assignment Data, PatentsView, SEC Edgar
ROA(t+1)	The ratio of operating income before depreciation to assets in year t+1.	Compustat
ROA(avg. t+1~t+3)	The average ratio of operating income before depreciation to assets from years t+1 to t+3.	Compustat
Cash Flow/Asset (t+1)	The ratio of operating activities net cash flow to assets in year t+1.	Compustat
Cash Flow/Asset(avg. t+1~t+3)	The ratio of operating activities net cash flow to assets from years t+1 to t+3.	Compustat
Independent Variables		
M	An indicator that takes on a value of one for the last month of a fiscal year (e.g., December for a 12/31 yearend firm).	Compustat
M-1	An indicator that takes on a value of one for one month prior to the last month of a fiscal year (e.g., November for a 12/31 yearend firm).	Compustat
M-2	An indicator that takes on a value of one for two months prior to the last month of a fiscal year (e.g., October for a 12/31 yearend firm).	Compustat
M-3	An indicator that takes on a value of one for three months prior to the last month of a fiscal year (e.g., September for a 12/31 yearend firm).	Compustat
M-4	An indicator that takes on a value of one for four months prior to the last month of a fiscal year (e.g., August for a 12/31 yearend firm).	Compustat
M-5	An indicator that takes on a value of one for five months prior to the last month of a fiscal year (e.g., July for a 12/31 yearend firm).	Compustat
Market Cap	The IHS-transformed market capitalization (number of outstanding stocks * stock price) in a firm-month	Compustat; CRSP
Monthly Returns	The buy-and-hold returns of the stock in a firm-month.	Compustat; CRSP
Patent Stock	The IHS-transformed total number of patents filed in the past twenty years plus the total number of net patent purchases over the past twenty years (i.e., the total number of purchased patents minus the total number of patents sold).	Patent Assignment Data; Kogan et al. (2017)
# Patents Granted	The IHS-transformed number of patents granted in a firm-month.	Kogan et al. (2017)

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Variable	Definition	Source
<i>High Core Patent Wedge</i>	A dummy variable takes on a value of one when difference between the percentage of patents sold in month M that are core and the exact same percentage computed across all other months of that fiscal year is in the highest quartile within year and zero otherwise.	Patentsview
<i>Low Core Patent Wedge</i>	A dummy variable takes on a value of one when difference between the percentage of patents sold in month M that are core and the exact same percentage computed across all other months of that fiscal year is in the lowest quartile within year and zero otherwise.	Patentsview
<u>Cross-Sectional Variables</u>		
<i>Earnings Forecast Surprise</i>	The difference between actual earnings and market expectations. Market expectations is measured as either the analyst consensus forecast if available or prior year's earnings if the firm does not have analyst coverage.	Compustat; I/B/E/S
<i>Meet or Just Beat (Consensus Forecast)</i>	An indicator variable that takes on a value of one when <i>Earnings Forecast Surprise</i> is between 0 and 1 cent.	Compustat; I/B/E/S
<i>Meet or Just Beat (2 Months Prior Forecast)</i>	An indicator variable that takes on a value of one when the difference between actual earnings and market expectation is between 0 and 1 cent. Market expectations is the consensus analyst forecast as of the second month of the last quarter of each fiscal year, following the Bhojraj et al. (2009).	Compustat; I/B/E/S
<i>High Sensitivity to Earnings and Stock Price</i>	A dummy variable takes on a value of one if a firm's total sensitivity to earnings and stock prices is in the highest quartile and zero otherwise. We measure the sensitivity by regressing executives' cash compensation on changes in ROA and stock returns in each Fama French 12 industry. The sum of the coefficients on changes in ROA and returns measures the direct and indirect effect of earnings' impact on executives' compensation (Leone et al. (2006)).	Compustat
<i>Private Firm</i>	A dummy variable takes on a value of one if the firm is a private firm and zero otherwise.	Patent Assignment Data
<i>High Financial Constraints (Hoberg and Maksimovic (2015))</i>	A dummy variable equal to 1 if Hoberg and Maksimovic (2015) measure is the highest quartile. Hoberg and Maksimovic (2015) measure: firms with higher values are more similar to a set of firms known to be at risk of delaying their investments due to issues with liquidity.	Compustat Hoberg and Maksimovic (2015)
<i>High Financial Constraints (Whited and Wu (2006))</i>	A dummy variable equal to 1 if the Whited and Wu (2006) index is in the highest quartile. Whited and Wu (2006) Index: $-0.091CFit - 0.062DIVPosit + 0.021TLTDit - 0.044LNTAit + 0.102ISGit - 0.035SSGit$	Compustat Whited and Wu (2006)

Notes: All continuous variables are winsorized at 1 % and 99 % to reduce the influence of outliers.

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